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5.7. Problems

- 1. Suppose that we have numerical representations $(V_1, \dots, V_I)$ for the preferences $(\succ_i)$ of our I consumers, and we are using a weighted utilitarian social welfare functional for some given collection of weights (α_i) to make social choices. Show that we can change to the (equal-weighted) utilitarian social welfare functional without changing the choices that are made by changing at the same time the numerical representations of the preferences $(\succ_i)$.
- 2. Consider the following social choice problem in the setting of consumption of two goods by two consumers. The two goods are called tillip and quillip and the two consumers are called 1 and 2. Consumer 1 has utility function $U_1(t, q) = 6 + .4 \ln(t) + .6 \ln(q)$ (where t is the amount of tillip 1 consumes and q is the amount of quillip). Consumer 2 has utility function $U_2(t, q) = 8 + \ln(t) + \ln(q)$. The social endowment consists of 15 units of tillip and 20 units of quillip.
 - (a) Suppose that a social dictator has social welfare functional of the following form: Social welfare, as a function of (u^1, u^2) is a weighted sum with weight 2 on the lesser of u^1 and u^2 and weight 1 on the greater of the two. What will be the welfare optimum plan chosen by this social planner?

5.7. Problems

- (b) What is the set of all feasible, Pareto efficient allocations of the consumption good for this society?

- 3. Prove proposition 5.4.
- 4. Construct two nonintersecting convex sets in R^2 such that the only separating hyperplane between them necessarily intersects each of the sets.
- 5. Suppose that in the setting of syndicate theory we have one risk neutral consumer and $I - 1$ who are strictly risk averse. As in the text, let the risk neutral consumer have index 1. We know (from fact 4) that the risk averse consumers' parts of any efficient sharing rule consist of a constant payment regardless of the state. The size of these payments depend, of course, on all the data in the problem; in particular, they depend on the weights of the consumers in the Bergsonian social welfare functional. Holding fixed everything else (that is, the utility functions of the consumers and the payoffs in each state of nature), write $y_i(\alpha_1, \dots, \alpha_I)$ for the payment to consumer i (for $i \geq 2$) as a function of the vector of weights. What can you say about how the function y_i changes as we change any of the individual weights?
- 6. Prove the fifth fact from the subsection on syndicate theory. The key is to find out the right values for the constants k_i . The k_i are defined so that the first-order conditions all hold and the adding up condition holds. There are $I + N$ unknowns (each value of k_i and each multiplier μ_n) and $IN + 1$ equations (IN first-order conditions and one adding up constraint), so you can see that some of the equations must be redundant, which is why this is such a special case.
- 7. In the context of the fifth fact concerning syndicate theory, suppose the syndicate has available a number of gambles that it might undertake. Specifically, for $m = 1, \dots, M$ the syndicate might undertake a project whose payoff in state s_n is Y_{mn} . The syndicate can undertake only one of these M alternative projects, it must undertake one, and the decision which to undertake must be made before any uncertainty concerning the state of nature resolves. In this case, the set of feasible social outcomes consists of sharing rules (y_{in}) such that for some $m = 1, \dots, M$, $\sum_{i=1}^I y_{in} = Y_{mn}$ for all n . What now is the set of efficient sharing rules? (Because of the discrete nature of the variable m , you seemingly cannot solve this problem just with calculus techniques. But consider expanding the range of feasible social outcomes still further to include the possibility that the choice of gamble to take may be made randomly.)

- 8. For each of the three examples of social choice rules given in section 5.4 (that is, the Pareto rule, majority rule, and the Borda rule), which of the four properties in Arrow's theorem does the rule obey, and which does it violate.
- 9. Suppose that we changed the definition of a dictator in property 4 to be: i is a dictator (in a given social choice rule) if $\succ_i \equiv \succ^*$ for any array of consumer preferences. That is, we add to the definition: When i is indifferent, society is indifferent. Prove that if property 4 is changed in this fashion, it is possible to find a social choice rule that satisfies the four properties. (If you like a challenge: How precisely can you categorize all social choice rules that obey the first three properties? We know there must be a dictator in the sense of the original property 4. But what can you say about cases in which this dictator is indifferent between two outcomes?)

part II

The price mechanism

chapter six

Pure exchange and general equilibrium

Prologue to part II

Part II of this book concerns the classic models of price-mediated exchange in markets. We begin in this chapter with situations of *pure exchange*, where consumers wish to exchange bundles of goods they hold at the outset for other bundles they will subsequently consume. We will use this setting to introduce the reader to the theory of *price-mediated market transactions* and, more particularly, the theory of *general equilibrium*, in which all markets in all goods are considered simultaneously. Then, for the rest of part II, *firms* are brought into the story. Chapter 7 will give some of the fundamental concepts of the classical theory of the firm, which is very much parallel to the theory of the consumer from chapter 2. Chapter 8 will consider, both in settings of partial and general equilibrium, models of competitive firms and markets. In chapter 9 we will introduce monopoly and in chapter 10 oligopoly.

Beyond treating these fundamental pieces of classical microeconomics, we will return throughout to a particular message: Price-mediated exchange in the marketplace and, in particular, the equilibrium notions that we use are “reduced forms” for some unspecified institutional mechanisms of exchange. Especially as we encounter the models of imperfectly competitive markets in chapters 9 and 10, questions will arise about why (and whether) these are appropriate “reduced forms,” questions that seemingly can be resolved only if we are more specific about the institutional mechanisms involved.

6.1. Pure exchange and price equilibrium

The problem

We imagine a world with a finite number I of consumers. The I consumers will be indexed $i = 1, \dots, I$. These consumers will all (shortly) sit

down to a grand feast at which they will do all their consuming. There are K commodities, indexed by $k = 1, \dots, K$, that the consumers might consume in varying amounts. Thus a consumption bundle for any of the I consumers is a vector x from R^K . We will assume that consumers cannot consume negative amounts of any of the commodities, but they can consume any nonnegative vector x from R^K , so we restrict attention to consumption bundles from the positive orthant of R^K , which we denote by X . Each of our I consumers has preferences over what she will consume; the preferences of consumer i are assumed to be represented (in the style of chapter 2) by the utility function $U_i : X \rightarrow R$. We assume that the consumers are all *locally insatiable* and that none of the goods is noxious, so that preferences (and the utility functions U_i) are nondecreasing. We assume that each consumer has continuous preferences.

We also imagine that these consumers come with *endowments* in the form of a commodity bundle from X . We let $e^i \in X$ represent the endowment of consumer i .¹

Our consumers could simply sit down and consume their endowments. But one consumer might, for example, be endowed with a lot of some good that she is not particularly fond of. She may wish to exchange some of that good for something that she likes more. So we wonder, Which exchanges will take place? What will each consumer wind up consuming?

The problem is one of *pure exchange* because everything that is eventually consumed comes from someone's endowment. If we write $e = \sum_{i=1}^I e^i$, then e is the total endowment of all the consumers in this society, and what consumers wind up consuming cannot exceed e in sum.

In the case of two consumers, we can use an Edgeworth box to depict the situation. Create an Edgeworth box as in the last chapter, where the origin of consumer 2's coordinate system is placed at the point $e = e^1 + e^2$ of consumer 1's coordinate system. If we mark point e^1 in 1's coordinate system (which is also point e^2 in 2's coordinate system), we get the picture in figure 6.1. Note in particular the shaded "lens" and the heavily marked portion of the contract curve. The lens represents all reallocations of the social endowment between the two consumers in a fashion that leaves them both at least as well off as if they consume their respective endowments. And the shaded portion of the contract curve represents all the Pareto efficient points that make one or the other or both better off than at their respective endowments.

¹ We use a superscript i so that we can write things like e_k^i for the amount of good k with which consumer i is endowed.

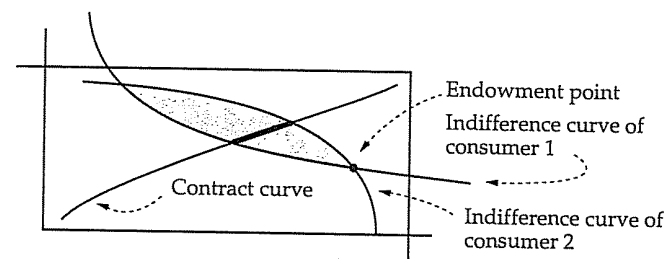


Figure 6.1. The Edgeworth box with endowments.

We show here an Edgeworth box with the endowment point of the two consumers marked. The shaded lens gives all reallocations of the social endowment (that are nonwasteful) that give each consumer at least as much utility as she obtains at her original endowment. And the heavy part of the contract curve represents all efficient reallocations with this property.

We call a collection of the objects above — a list of consumption goods, a list of consumers, and utility functions and endowments for each consumer — a *pure exchange economy*.

Price equilibrium

Now imagine that these goods are traded in markets, where p_k is the price of good k , and $p = (p_1, \dots, p_K)$ is used to denote the *price vector*. Imagine that each of our I consumers is aware of the prices given by p , and each believes that she cannot by her actions affect any of these prices. Then consumer i has at her disposal wealth $Y^i = p \cdot e^i$. That is, she can take her endowment to each market and sell it. She will receive $p_k e_k^i$ for her endowment of good k ; summing over all the k yields Y^i . She can then spend this wealth buying whatever bundle x^i maximizes her utility, subject to her budget constraint $p \cdot x^i \leq Y^i$. Of course, if she wishes to save on the number of trips she has to make, she can do the whole thing in one go-round. Having worked out which bundle she wishes to consume, she first sells commodities for which her endowment exceeds her desired consumption and then buys more of the goods for which her endowment is less than desired.

However we imagine this happening, we have the following *problem of consumer i at prices p* :

$$\text{maximize } U_i(x) \text{ subject to } x \in X \text{ and } p \cdot x \leq p \cdot e^i. \quad \text{CP}(i, p)$$

Definition. A general equilibrium, also called a *Walrasian equilibrium*, for a given pure exchange economy consists of a price vector p and consumption bundles x^i for each consumer such that

(1) at prices p , x^i solves the problem $CP(i, p)$

(2) markets clear: $\sum_{i=1}^I x^i \leq \sum_{i=1}^I e^i$

The array of final consumption vectors (x^i) is referred to as the *final consumption allocation* in the equilibrium.

Following in the footsteps of generations of classical microeconomists, we make the rather heroic assertion that in many situations of pure exchange consumers will wind up at the consumption allocation part of some Walrasian equilibrium for the economy, and insofar as there are markets in these goods, prices will correspond to equilibrium prices.

If you weren't before, perhaps you are finally rather impressed with the chutzpah of generations of microeconomists. If so, understand that no one is asserting that in every situation of pure exchange the outcome will resemble constituent pieces of a Walrasian equilibrium. For example, with two consumers, the case of the Edgeworth box, we might assert (or at least hope) that the two would find their way to the contract curve, and we'd be rather surprised if one or the other wound up worse off than where she had started. But we wouldn't necessarily expect the outcome to correspond to a Walrasian equilibrium. The sense of most economists who believe in Walrasian equilibrium is that it takes "many" consumers, no one (or few) of whom hold monopoly positions in one or more of the goods, to believe that the previously made assertion is apt to be correct, and it takes an institutional framework in which something like prices appear — something that establishes the "rates" at which one commodity can be exchanged for another.

We will discuss in the next section (a) reasons to believe this assertion, (b) reasons given sometimes that do not inspire belief, and (c) some reasons for doubt. But first, we give a few simple technical notes.

(1) If we multiply all prices in a Walrasian equilibrium by a positive scalar, the solutions to the consumer's problems don't change, and so we still have an equilibrium. In other words, if $(p, (x^i))$ is a Walrasian equilibrium, so is $(\lambda p, (x^i))$, for any (strictly) positive scalar λ . Put still differently, what is important in a Walrasian equilibrium are the relative prices — the rates at which one commodity can be exchanged for another.

(2) We have assumed that consumers' preferences are nondecreasing. We can conclude from this assumption that every price p_k must be nonneg-

ative at any equilibrium. This is so because if $p_k < 0$, then a consumer could "buy" lots of good k at no decrease in her utility and "buying" a good with a negative price means more resources to buy other goods. We assume consumers are locally insatiable, so if some price were negative, consumers would never find an optimal solution to their consumption problem, and we certainly wouldn't find markets clearing in this good.^a

(3) Since we have assumed that consumers are locally insatiable, they are sure to spend all their wealth at any solution to $CP(i, p)$; that is, $p \cdot x^i = p \cdot e^i$. By summing over all consumers' budget constraints, we conclude that

$$\sum_i p \cdot x^i = \sum_i p \cdot e^i.$$

Yet we only require that $\sum_i x^i \leq \sum_i e^i$. By virtue of (2) we know that prices are nonnegative, so we conclude that the only goods k for which $\sum_i x_k^i < \sum_i e_k^i$ is possible are goods whose price p_k is zero. (Be sure you see why this follows from [2].)

(4) We could never have a Walrasian equilibrium where all prices are zero, as long as consumers (or even just one consumer) are locally insatiable.

(5) While we don't take quite seriously the notion of a Walrasian equilibrium in a two-person economy, nonetheless it may be instructive to see what these equilibria look like in an Edgeworth box. Refer to figure 6.2(a). We have put a negatively sloped line through the endowment point; the slope of this line represents a particular set of relative prices for the two goods, and the shaded triangle represents the set of budget-feasible consumption bundles for consumer 1 at these relative prices. Hence consumer 1, solving her problem at these relative prices, would select the point marked with a star. The crucial thing is that at these relative prices, the same "budget line" works for consumer 2 in his coordinate system. It should take you a moment's reflection to see why this is true. To help you see this, we have drawn the relative prices in figure 6.2(a,b,c) so that the price of good 1 is half that of good 2. Hence consumer 2's budget feasible set is the shaded region in figure 6.2(b), and he chooses the point marked

^a To come to this conclusion it would suffice to have one consumer who is locally insatiable and has nondecreasing preferences, since it is implicit in the definition of an equilibrium that every consumer's problem has a finite solution. For some goods, such as pollution, it may not be sensible to assume that preferences are nondecreasing. To allow for such goods, one must allow for negative prices and replace the inequality in the "markets clear" relationship with an equality. You will find this variation discussed in most books that study general equilibrium in detail.

with the box. If we superimpose the two, as we do in figure 6.2(c), we see that we don't have an equilibrium at these relative prices; consumers 1 and 2 together want more of the first good than is available.

In figure 6.2(d) we show what things look like at an equilibrium. At the relative prices shown there, both consumers choose the point marked with the star, although this star represents different bundles in the two different coordinate systems. Note carefully that each consumer's indifference curve through the point $\star$ is tangent to the budget line at that point, and hence the two indifference curves are tangent to each other at $\star$. That is, the Walrasian equilibrium allocation lies along the contract curve; it is Pareto efficient. This is a "proof in a picture" of a very general fact that we will prove in section 6.3.

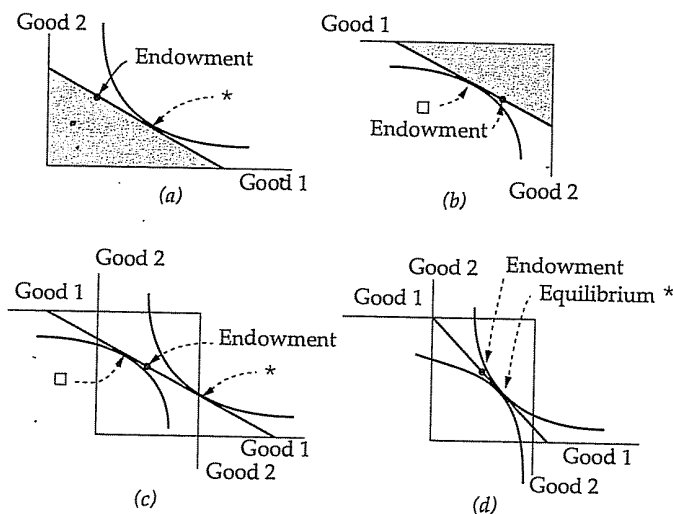


Figure 6.2. Walrasian equilibrium and the Edgeworth box.

In (a), we show the demand of consumer 1 at a particular set of relative prices, and in (b), we see consumer 2's demand at those prices. Superimposing the two in (c), we see that these demands are inconsistent; this is not a Walrasian equilibrium. In (d) we see the picture of a Walrasian equilibrium, and in particular we see that a Walrasian equilibrium lies along the contract curve.

6.2. Why (not) believe in Walrasian equilibrium?

Why should anyone believe that the outcome of exchange will look like a Walrasian equilibrium? Or, rather, in what situations would it be sensible to believe this assertion? One thinks first of a free-market economy in which goods do sell at some prices, where consumers know those prices, and where consumers can buy and sell commodities in whatever quantities they wish at the prevailing prices. If we ever found consumers in such a situation, then we would have consumers who are aware of the relative rates of exchange of each commodity for every other. If all consumers take these rates of exchange as given, if all consumers face the same rates of exchange, if all maximize their utility as in chapter 2, and if all consumers are in the end happy with the exchanges they have made, then we would necessarily have a Walrasian equilibrium. Let us make a list of some things this suggests we might worry about.

(1) Each consumer must be aware of all the prices for every good. In reality, some goods trade at more than one price in different locations. There are reasons that this could persist which are consistent with every consumer knowing all the prices at all locations, (such as buying a good at a particular location bundles together the purchase of the good with different levels of service, different ambience, and so on). But it seems unrealistic to suppose that all consumers know all prices in all locations; anyone who has ever shopped for a new car (at least, in the United States) will know this. As soon as consumers face different prices, the assertion is in trouble (unless we modify the definition to allow for prices that change with the individual consumer).

Another aspect concerns dynamic choice in an uncertain world. Insofar as some of the exchanges made today depend on exchanges the consumer hopes she will be able to make later, today's choices depend on forecasts of later prices and market opportunities. (Consider, for example, buying a particular car based on a hypothesis about the price of gasoline that will later prevail.) Do consumers accurately forecast all subsequent prices? And what if, a priori, there is uncertainty about the future that will later resolve, the resolution of which affects later prices? (For example, the price of gasoline three years hence may depend on the political situation in various parts of the world, oil reserves found or not found in particular locations, and so on.) Economists adapt the model of Walrasian equilibrium to situations where time and/or uncertainty play a role. These adaptations will be discussed briefly in section 6.5; until then (and, perhaps, even after), scepticism about how well the basic assertion will do when time and/or uncertainty plays a role seems warranted.

(2) Consumers must be able to buy or sell as much or as little as they want at the going prices. There are (at least) two reasons to doubt this. First, consumers might wish to transact in such large amounts that they affect the prices they face. J. D. Rockefeller, seeking to sell all the oil he owned, would probably have found that an attempt to do so would have changed the price of oil. The second reason for doubt is at a more pedestrian level. In many economies rationing and stock-outs are not unknown. In some economies, a consumer may wish to buy a leg of lamb and may know the "price" of lamb that is supposed to be charged, but there is no guarantee that any lamb will be in stock at the butcher shop.

(3) Consumers may not act rationally; that is, in the fashion of chapter 2. Little is left to say about this that we haven't already said.

(4) In most economies where we might hope to meet all the conditions mentioned, the economy is not one where agents simply exchange their endowments. In most economies production also takes place. If all the good candidate economies have production, then the foregoing definition, which involves pure exchange, may be descriptive of almost nothing.

We will introduce production in the context of private enterprise economies in the next few chapters, but for now let us mention a variation on the basic definition of a Walrasian equilibrium that will go at least partway toward meeting this source of worry. We imagine an economy with K commodities, I consumers, and a "productive sector" — either a sector of firms or a government enterprise sector or some mix. We can think of the total effect of this productive sector as: a certain vector of commodities z^I is bought as input and another vector z^O is sold as output. Then,

Definition. A Walrasian equilibrium for a given pure exchange economy with fixed total production input z^I and output z^O consists of a price vector p and consumption bundles x^i for each consumer such that the following hold:

(1) for each i , at prices p , x^i solves the problem $CP(i, p)$ and

(2) markets clear: $z^I + \sum_{i=1}^I x^i \leq z^O + \sum_{i=1}^I e^i$.

In other words, we take the production plans as exogenously given and look at the "consumer exchange" part of the economy. In later chapters we will move the production plans into the category of things endogenously chosen. But we could, with the definition above, at least study consumer exchange for each fixed production plan.^b

^b At the risk of being overly pedantic and confusing the reader, a caution should be entered here when you think of the productive sector as a state enterprise. In such cases, some of the inputs to production may result from taxation of consumers' initial endowments, or from taxes on the sale and purchase of particular items, especially labor services. The consumer's problem would have to be changed to reflect any such taxation.

For each of the reasons given above and others besides, you might not think that the results of trading, even in a well-functioning market economy, would result in a Walrasian equilibrium. To deal with these criticisms, many variations on the basic model have been created. For example, there is an enormous literature (to which you will shortly be introduced) about price equilibrium in markets where time and/or uncertainty play a role. There are attempts to model equilibrium price dispersion (caused by lack of perfect information). There is analysis of situations where some consumers can change prices with their actions (although it is far more typical to think that firms have this sort of power; we will see such models in chapters 9 and 10). There is a literature on equilibrium where consumers can be rationed. All these are variations on the theme of Walrasian equilibrium as defined above. So both to understand situations where these qualifications aren't needed and to set a benchmark against which such variations can be measured, we proceed with our study of Walrasian equilibrium.

Mechanisms behind Walrasian equilibrium

While we may look at market economies for cases in which Walrasian equilibrium is a reasonable solution concept, one thing that the concept of a Walrasian equilibrium doesn't provide is any sense of how markets operate. There is no model here of who sets prices, or what gets exchanged for what, when, and where. One thing that we see in most market economies and that is completely missing from the story is *money*. (Prices have to be measured in some numeraire but, as we have already seen, in the theory it is only relative prices of commodities that matter.) Because of this, a Walrasian equilibrium is a *reduced form* solution concept; it describes what we imagine will be the outcome of some underlying and unmodeled process. It seems natural to think that we could increase (or decrease) our faith in the concept of a Walrasian equilibrium if we had some sense of how markets really do operate. And economists have studied specific models of market/exchange mechanisms, attempting to see whether those mechanisms confirm Walrasian equilibrium as an appropriate reduced form solution concept.

A fairly unrealistic mechanism, which nonetheless populates some of the literature on general equilibrium, concerns an individual known as the Walrasian auctioneer. This individual stands up in front of the entire population of an economy and calls out a price vector p . Every consumer consults her preferences and determines what she would want to buy and sell at these prices; we will write $z^i(p) = x^i(p) - e^i$ for the *net trade* that consumer i desires at prices p . Each consumer reports back to the

auctioneer what net trade she wishes, and the auctioneer adds them up; if $\sum_{i=1}^I z^i(p) \leq 0$, everyone breathes a sigh of relief and the net trades are executed. But if markets don't clear at prices p , then the auctioneer tries a different price vector p' , and so on. One can pose various rules that the auctioneer might employ in determining what prices p' to quote if p doesn't work — say he raises the price of goods that are overdemanded (that is, any good k for which $\sum_{i=1}^I z_k^i(p) > 0$) and lowers the price of those that are underdemanded. We may wonder whether this scheme of price adjustment will lead to an equilibrium. You can find discussion of this in the literature; look for references to the *tatonnement* adjustment process. (We won't carry further on this subject because Walrasian auctioneers are not usually found in real economies.)

A second unrealistic mechanism involves each consumer simply reporting to the central authorities her desired excess demands at every price p . (In cases where at a price vector p the consumer is indifferent between several excess demand vectors, we will allow her to submit all the alternatives she would be happy to take.) The authorities (who previously employed the Walrasian auctioneer) examine all these submissions and look for one or more prices p where markets clear. (We won't say what they do if the excess demand functions submitted are not consistent with market clearing at any price.)

A major reason why the two previous mechanisms seem unrealistic is that they involve equilibration of all markets for all consumers at once. It is more realistic to think of institutions and mechanisms where either each market operates separately or trades are made between pairs of individuals.

For example, we can imagine consumers wandering around a large market square, with all their possessions on their backs. They have chance meetings with each other, and when two consumers meet, they examine what each has offer, to see if they can arrange a mutually agreeable trade.² To be very precise, we might imagine that at every chance meeting of this sort the two flip a coin and depending on the outcome one is allowed to propose an exchange, which the other must either accept or reject. If an exchange is made, the two swap goods and wander around in search of more advantageous trades made at chance meetings. The rule is that you can't eat until you leave the market square, so consumers wait until they are satisfied with what they possess and then leave to have a picnic in the adjacent park. In this story, a smart consumer who hates

² For those readers who follow American baseball, think of general managers trading players, but on a larger scale.

apples may nonetheless give up some of her pears for someone else's apples if the apples are offered on good terms, in the speculative hope that she will later be able to trade the apples for the cherries that she really wants.³

Or we can imagine models in which it is imagined that markets in the various goods operate independently. One might imagine a model with market makers — individuals who specialize in a particular commodity or related commodities. Over in one corner of the market square is the cheese vendor who is happy to buy or sell cheese (for money or for some standard good, such as wheat or gold), in another corner is a specialist in corn, and so on. We might imagine that the cheese vendor posts prices at his stall, in which case we might want to have a few cheese vendors with stalls adjacent to each other to provide some competition, or we might imagine more traditional bargaining between anyone who comes up to the cheese stand and the cheese vendor. The point is that all transactions in cheese go via the cheese vendor; if consumer 1 is endowed with too much cheese and consumer 2 with too little, consumer 1 sells her excess cheese to the cheese vendor, who then resells some of this to consumer 2.

Or we could organize the cheese market without a specialist cheese vendor. Making, say, wheat the medium of exchange, we might imagine that at 10:05 A.M., it is time to auction off Stilton cheese. Anyone with Stilton to sell is allowed to post an "asking price"; this person will sell a standardized amount of Stilton for so many bushels of wheat. At the same time, those interested in buying Stilton are allowed to post "bids" — so many bushels of wheat for a piece of Stilton. At any point in time, a consumer can "take" any offer that is currently posted: Consumers can post new offers (on either side of the market), they can improve offers they already have posted, and they can retract offers they previously have made. The posting of offers can be done by open outcry or by some fancy electronic posting service. (Consumers all sit at CRT terminals and can enter bids or asks; the CRT always displays the current "best" bids and asks; and one can take the best posted offer at the touch of a particular key on the keyboard.) All this takes place from 10:05 until 10:20, after which it is time for Camembert, and so on.

The exploration of more realistic models of markets is in relative infancy. A large part of this literature appears in the monetary economics (the economic theory of money) and is concerned first and foremost with the use of money in these types of markets. The analysis of the "random

³ Fans of baseball may recall the occasional trade based on this sort of calculation.

meetings" marketplace has been quite popular recently, and the analysis that has been done offers *qualified* support for the Walrasian equilibrium notion. Analysis of markets in a single commodity, both of the market-maker and two-sided auction variety (as well as one-sided auctions) can be found as well, although little of this work has extended from the case of a single market to issues associated with general equilibrium. References will be provided in section 6.6, although in general you will need to get through part III of this book before this literature will be accessible.

Experimental evidence

If you consult these references, you will find that the analysis is hard and the conclusions require consumers who are extraordinarily sophisticated. You might worry, in consequence, that the results derived in support of the concept of Walrasian equilibrium are not very credible. To assuage these worries, you should consult the literature on experimental economics. A number of researchers have attempted to see whether markets perform under controlled conditions in the way economists assume they do. In the typical experiment, different subjects are induced to have different valuations for some good, relative to a numeraire good (money). Some of the subjects who are initially endowed with the good put a relatively low value on it; others who do not have it put a relatively high value on it. From the data of the experimental conditions, one can compute what the "equilibrium price" for the good should be (if standard microeconomic theory is correct). The subjects are then put into an exchange environment, to see whether they make exchanges the way the theory says they will. The CRT-terminal-mediated double "oral" auction market mechanism is often used.

The results obtained are usually striking in their support of Walrasian equilibrium. Subjects are not given the data needed to calculate a priori what the equilibrium prices "ought" to be. So the first time the market operates, some trades are made far from the theoretical equilibrium. But if the market is repeated (endowments are restored), subjects typically learn extremely quickly what the market clearing price for the good will be, and they trade in a way that is quite close to what theory predicts.

These experiments do not quite get to the level of generality of a Walrasian equilibrium for a general equilibrium with many interdependent markets. But the repertory of experiments is growing quickly and, except for a few special cases, those experiments that have been run are consistent with the notion of Walrasian equilibrium. All in all, they make Walrasian equilibrium look quite good.

6.3. The efficiency of a general equilibrium

We now turn to properties of Walrasian equilibrium. We return to the setting of section 6.1 with K commodities and I consumers who exchange those commodities. We suppose that a Walrasian equilibrium is achieved. And we ask: Is this Walrasian equilibrium a *good* way to allocate resources in this economy? Putting it in slightly melodramatic and ideological terms, can we confiscate everyone's endowment, reallocate goods, and in so doing obtain an allocation better than that achieved in a Walrasian equilibrium?

Of course, to answer this question we need a sense of what we mean by "better." Following developments from the last chapter, we use the concept of Pareto efficiency.

The two theorems of welfare economics

Having chosen Pareto efficiency as our criterion, we can recall a remark made in section 6.1 about the Edgeworth box: In a two-person, two-good economy, the allocation part of a Walrasian equilibrium is Pareto efficient (that is, lies along the contract curve). This result generalizes very neatly.

Theorem. The First Theorem of Welfare Economics. *A Walrasian equilibrium always yields a Pareto efficient allocation of the social endowment.*⁴

The proof is quite simple. Suppose that $(p, (\hat{x}^i))$ is a Walrasian equilibrium and $(\tilde{x}^i)$ is a Pareto superior allocation of the social endowment. We will derive a contradiction.

Since $(\tilde{x}^i)$ is a reallocation of the social endowment $e = \sum_{i=1}^I e^i$, we have $\sum_{i=1}^I \tilde{x}^i \leq e$. Since equilibrium prices are nonnegative, this implies

$$\sum_{i=1}^I p \cdot \tilde{x}^i \leq p \cdot e. \quad (*)$$

Since preferences are locally insatiable, Walras' law holds with equality for each consumer and summing these equations yields

$$\sum_{i=1}^I p \cdot \hat{x}^i = p \cdot e. \quad (**)$$

⁴ We maintain here the earlier hypotheses that consumers are all locally insatiable and have nondecreasing preferences. We wouldn't need to assume nondecreasing preferences if we posed the "markets clear" conditions in the definitions of an equilibrium and a feasible allocation as equalities.

By assumption, each consumer i likes $\tilde{x}^i$ as much as $\hat{x}^i$, and some one (or more) consumer likes $\tilde{x}^i$ more. Consider consumer i . Since this consumer at least weakly prefers $\tilde{x}^i$ to $\hat{x}^i$, it must be that $p \cdot \tilde{x}^i \geq p \cdot \hat{x}^i = p \cdot e^i$. For if $p \cdot \tilde{x}^i < p \cdot \hat{x}^i$, then by local insatiability i could afford something strictly better than $\tilde{x}^i$ at prices p , which would be strictly better than $\hat{x}^i$, which would then contradict the supposed optimality of $\hat{x}^i$ for i at prices p . Similarly (in fact, even more simply), for any consumer i such that $\tilde{x}^i$ is strictly better than $\hat{x}^i$, we must have $p \cdot \tilde{x}^i > p \cdot \hat{x}^i = p \cdot e^i$. This must hold for at least one consumer since we assume $(\tilde{x}^i)$ Pareto dominates $(\hat{x}^i)$, and so by summing up these weak and strict inequalities we obtain

$$\sum_{i=1}^I p \cdot \tilde{x}^i > \sum_{i=1}^I p \cdot \hat{x}^i.$$

Comparing this inequality with (*) and (**) gives us the desired contradiction and completes the proof.

You should now be hearing choirs of angels and choruses of trumpets. The "invisible hand" of the price mechanism produces equilibria that cannot be improved upon.

Of course, one can wonder whether the price mechanism produces any equilibrium at all; the result above is predicated on the existence of a Walrasian equilibrium. We will attend to questions about the existence of equilibrium in the next section. But granting existence, it should still be noted that "improved upon" in the previous paragraph refers to Pareto improvements. There is no reason to suppose that the equilibrium allocation is not vastly inequitable, giving lots of utility to some consumers and very little to others. In response to this consideration, we have the following result.

Theorem. The Second Theorem of Welfare Economics. Assume that preferences are convex, continuous, nondecreasing, and locally insatiable. Let $(\hat{x}^i)$ be a Pareto efficient allocation of the social endowment that is strictly positive: $\hat{x}_k^i > 0$ for all i and k . Then $(\hat{x}^i)$ is the allocation part of a Walrasian equilibrium, if we first appropriately redistribute endowments among the consumers.

In other words, if one imagines that it would be difficult for a social dictator to find an equitable and efficient allocation of the social endowment, and if the economy has well-functioning markets, the dictator might choose to reallocate initial endowments in an equitable fashion and then let the market take over. If you wonder how the social dictator could decide a

priori on an equitable initial redistribution of the social endowment, see problem 3 for at least one idea.

Here is a sketch of the proof. Suppose $(\hat{x}^i)$ is the Pareto efficient allocation of the social allocation e that we wish to obtain as a Walrasian equilibrium. Let $\hat{e} = \sum_{i=1}^I \hat{x}^i$. Note that $\hat{e} \leq e$; a strict inequality is possible for some components. (We will have to work around this possibility.)

Define

$$X^* = \{x \in X : x \text{ can be allocated among the } I \text{ consumers in a way that strictly Pareto dominates } (\hat{x}^i)\}.$$

We assert (and you can prove for homework) that the set X^* is convex because preferences are convex. Let

$$X^\dagger = \{x \in R^K : x \leq e\}.$$

This is obviously a convex set.

The supposed Pareto efficiency of the "hat" allocation ensures that X^* and $X^\dagger$ do not intersect. Thus there is a separating hyperplane between them — a nonzero K -dimensional vector $p = (p_1, \dots, p_K)$ and a scalar β such that $p \cdot x \leq \beta$ for all $x \in X^\dagger$ and $p \cdot x \geq \beta$ for all $x \in X^*$. By local insatiability and continuity, we can show that $\hat{e}$, which is in $X^\dagger$, is on the boundary of X^* , and so $p \cdot \hat{e} = \beta$. We assert that $p \geq 0$; the proof is left to you. (Use the shape of the set $X^\dagger$.) We assert that $p \cdot e = \beta$ as well; since preferences are nondecreasing we can take any goods left over from the social endowment in the allocation $(\hat{x}^i)$ and allocate them among the consumers and then repeat the argument using local insatiability and continuity. Thus we know that if $\hat{e}_k < e_k$ for some good k , then $p_k = 0$.

Let $\hat{e}^i$ be any allocation of the social endowment e such that $p \cdot \hat{e}^i = p \cdot \hat{x}^i$. We know that such allocations exist because the goods not fully used up in the allocation $(\hat{x}^i)$ must have zero price.

We claim that $(p, (\hat{e}^i))$ is a Walrasian equilibrium for the economy if consumers start out with the endowments $(\hat{e}^i)$. To see this, suppose it is not a Walrasian equilibrium. Partition the set of consumers into two subsets: those for whom $\hat{x}^i$ is the best they can afford at prices p with endowment $\hat{e}^i$ and those for whom there is something better they can afford. Since by assumption $(p, (\hat{x}^i))$ is not a Walrasian equilibrium, the second set in this partition is nonempty.

For each i from the second set in the partition, let x^i be any bundle which i strictly prefers to $\hat{x}^i$ and that is affordable at prices p with endowment $\hat{e}^i$. By continuity of preferences, for some positive scalar $\alpha^i < 1$, the bundle $\alpha^i x^i$ is strictly preferred to $\hat{x}^i$ and satisfies $p \cdot \alpha^i x^i < p \cdot x^i \leq p \cdot \hat{e}^i$.^c

^c This implicitly assumes that $p \cdot x^i > 0$. But if $p \cdot x^i = 0$, since $\hat{x}^i$ is strictly positive, $p \cdot \hat{e}^i = p \cdot \hat{x}^i > 0$. And then $p \cdot x^i < p \cdot \hat{e}^i$, which is what we really need.

Define $\tilde{x}^i = \alpha x^i$. Let $s = \sum p \cdot (\hat{x}^i - \tilde{x}^i)$ where this sum is over the second set of consumers. By construction $s > 0$. Using this "budget surplus" and the local insatiability of preferences, it is possible to find bundles $\tilde{x}^i$ for the consumers i in the first group such that $\tilde{x}^i$ is strictly preferred by i to $\hat{x}^i$ and such that $\sum_{i=1}^I p \cdot \tilde{x}^i < \sum_{i=1}^I p \cdot \hat{x}^i = \beta$. But since $(\hat{x}^i)$ is strictly Pareto superior to $(\tilde{x}^i)$, $\sum_{i=1}^I \tilde{x}^i \in X^*$. And for all $x \in X^*$, $p \cdot x \geq \beta$. This gives the desired contradiction, completing the proof.

There are two remarks to make about this theorem and proof. First, the assumption that $(\hat{x}^i)$ is a strictly positive allocation is much too strong. There are, presumably, goods that one consumer doesn't like at all; these goods don't lower her utility (we're assuming nondecreasing preferences), but they do nothing to raise her utility. If these goods are appreciated by others, then strict positivity of the allocation is inconsistent with Pareto efficiency! To repair this catastrophe, there are numerous ways to proceed. One of the simplest is to say that a good is *desirable* if everyone's preferences are strictly increasing in that good. Then we could replace the assumption that $(\hat{x}^i)$ is strictly positive with the much less obnoxious assumption that every consumer is given, in the allocation $(\hat{x}^i)$, a positive amount of at least one desirable good. You are asked in problem 4(c) to show that this will be enough to get the result of the second welfare theorem.

Second, we used a separating hyperplane argument in the previous chapter to prove proposition 5.2, and it may be useful to highlight the differences. Here, we are separating two convex sets in the space of commodity bundles, R^K . Separating hyperplanes in this space have the right dimensionality to be prices. In proposition 5.2, we were separating convex sets in the space of utility imputations, R^I , and the hyperplane gave us "weights" to put on the utilities of the various consumers. Note also that in proposition 5.2, we obtained convexity of one of the two separated sets from an assumption that the utility functions were concave functions. (To be precise, in the proof of proposition 5.2, we assumed convexity by fiat, by looking at the convex hull of $V(X^i)$. But we justified this approach in proposition 5.3 by assuming that utility functions were concave.) Here we are making the weaker assumption that preferences are convex; we don't need to assume that they have a concave numerical representation.

Externalities

Do you hear the angels and trumpets now? There are good reasons still not to hear them having to do with all the hidden assumptions made in the story of Walrasian equilibrium. We will discuss one of these, *externalities*, in some detail here.

In the model of an exchange economy posed above, each consumer is an "island" given prices. In particular, the utility of each consumer is unaffected by the consumption decisions of other consumers, if prices are held fixed.

In reality, matters may be more complex. Consumer i may enjoy filling his front yard with junk or may wish to add to the traffic of a

particular highway, which will depress the utility of neighbor i' and fellow driver i'' . On the other hand, consumer j may take good care of her garden, which raises the utility level of neighbor j' .

These are simple examples of *externalities*, situations where the consumption of some good(s) by one consumer affects the utility of another. In the first two cases, consumer i is generating negative externalities for i' and for i'' . In the third case, consumer j is providing positive externalities for j' . When there are externalities, we can't speak of the utility of consumer i as being represented by a function $U_i : X \rightarrow R$; now the domain for i 's utility function will potentially be the *entire* vector of consumption $x = (x^1, x^2, \dots, x^I) \in X^I$. Following the notation of chapter 5, we will use V_i to represent the preferences of consumer i over entire consumption allocations.

What do we mean by a Walrasian equilibrium when there are externalities? To see why there is a question here, imagine that consumer 1 attaches greater value to keeping her yard in good shape if her neighbor, consumer 2, keeps his yard clean. (To be precise, assume that the marginal utility consumer 1 attaches to keeping her yard neat increases with increases in the neatness of 2's yard.) Then at any prices, the amount of money consumer 1 will wish to spend on her yard depends on how neatly she believes her neighbor will keep his. If she believes that he will keep his yard in good shape, she will wish to spend more than if she believes that he will keep his yard in poor shape. We can't tell how much she will wish to spend if we don't know what he will spend on his yard. And, assuming the externality works in both directions, we can't say what he will spend until we know what she is going to spend. There is a simultaneity problem involving the two utility maximizations.

It is typical when modeling general equilibrium with externalities to come to grips with this by supposing that each consumer chooses her own consumption levels taking as given the *equilibrium* consumption levels of her fellow consumers. We define:

Definition. A Walrasian equilibrium with externalities consists of prices p and a consumption vector $\hat{x} = (\hat{x}^1, \dots, \hat{x}^I)$ such that

(a) For each consumer i , $\hat{x}^i$ is the solution in the variable x^i to the problem

$$\begin{aligned} &\text{maximize } V_i(\hat{x}^1, \dots, \hat{x}^{i-1}, x^i, \hat{x}^{i+1}, \dots, \hat{x}^I) \\ &\text{subject to } x^i \in X \text{ and } p \cdot x^i \leq p \cdot e^i. \end{aligned}$$

(b) Markets clear: $\sum_{i=1}^I x^i \leq \sum_{i=1}^I e^i$.

We might wonder whether this is the appropriate definition: Will consumers in an exchange economy in which there are externalities find their way to a Walrasian equilibrium?⁵ Setting that question aside, we can ask if Walrasian equilibria in this setting will be efficient.

The answer is in general no. If there are externalities, our proof of the first welfare theorem will not work. Specifically, the proof breaks down at the step where we say that if $(\tilde{x}^i)$ is Pareto superior to $(\hat{x}^i)$, then $p \cdot \tilde{x}^i \geq p \cdot \hat{x}^i$, with strict inequality for some consumers i . Before, $(\tilde{x}^i)$ being Pareto superior to $(\hat{x}^i)$ meant that consumer i would be at least as well off consuming $\tilde{x}^i$ as $\hat{x}^i$, *no matter what her fellow consumers did*, because her utility only depended on her own consumption. But to say that $(\tilde{x}^i)$ is better for i than $(\hat{x}^i)$ does not imply that

$$V_i(\hat{x}^1, \dots, \hat{x}^{i-1}, \tilde{x}^i, \hat{x}^{i+1}, \dots, \hat{x}^I) > V_i(\hat{x}^1, \dots, \hat{x}^{i-1}, \hat{x}^i, \hat{x}^{i+1}, \dots, \hat{x}^I).$$

That is, consumer i cannot necessarily do better (staying within her budget set) by changing only her own level of consumption. The components of $(\tilde{x}^i)$ might all be affordable, and $(\tilde{x}^i)$ might be Pareto superior, and still the components of $(\hat{x}^i)$ might solve the individual consumers' maximization problems *given* other consumers are choosing their parts of $(\hat{x}^i)$.

Establishing that the proof breaks down is not the same thing as showing that equilibrium allocations will be inefficient. We will offer a loose argument in a particular setting that every equilibrium allocation will be inefficient owing to externalities.⁶ Since this argument applies to every equilibrium allocation, it shows that both the first and second theorems aren't valid when there are externalities. (The loose argument that follows requires that you have at least a sense of Taylor series expansions. If this sort of thing is entirely foreign to you, then skip the next paragraph.)

Imagine a three-person economy in which the first commodity is "gardening services," the consumption of which makes one's yard more beautiful. Imagine that two of the consumers in this society live in adjacent houses, while the third lives on the other side of a particularly large mountain. Consumption by the third consumer of gardening services generates no externality for the other consumers, but each of the others generates

⁵ Thinking ahead to part III of the book, we can suggest why they might not, at least in some cases. Suppose that in equilibrium i and i' spend very little on their respective yards, and suppose each would happily spend more if the other spent more. Then we might imagine a neighborhood conference where each promises to spend more on condition that the other spends more. We have to worry about how this would be enforced; but in chapter 14 we will see one quite natural enforcement mechanism for this sort of agreement.

⁶ In problem 5 you are asked to provide a precise example to go with this argument, and in problem 6 you are given another concrete example to work through.

a positive externality for her neighbor through the consumption of gardening services. At any prices p , each consumer chooses the level of gardening services to consume in a way that equates the ratio of marginal utilities of gardening services to the marginal utility of consumption of other goods to the ratio of their prices. That is, each sets

$$\frac{\partial V_i}{\partial x_1^i} / \frac{\partial V_i}{\partial x_2^i} = \frac{p_1}{p_2},$$

where x_1^i is the level of consumption of gardening services by i , and x_2^i is i 's level of consumption of some other good that doesn't generate any externalities. Thus if we increase the amount of gardening services that, say, consumer 1 consumes by some small amount δ and lower the amount of good 2 that she consumes by $(p_1/p_2)\delta$, the marginal impact on her utility is approximately

$$\frac{\partial V_1}{\partial x_1^1} \delta - \frac{\partial V_1}{\partial x_2^1} \frac{p_1}{p_2} \delta = 0.$$

Similarly, if we simultaneously decrease the amount of gardening services that consumer 3 consumes by δ and raise the amount of good 2 that he consumes by $(p_1/p_2)\delta$, the impact on his utility is approximately zero. So this small reallocation doesn't change the utilities of either consumer 1 or consumer 3. But because of the positive externality that raising the level of gardening services consumed by consumer 1 has on the utility of consumer 2, this small reallocation improves the utility of consumer 2. Thus this small reallocation is Pareto improving, showing that a Walrasian equilibrium will not be Pareto efficient.

We will return to externalities in chapter 8; there are other interesting examples to discuss, but they require an economy with a productive sector. For now, we simply note that externalities give one reason to doubt that markets lead to efficient allocations. There are other reasons to doubt the conclusions of the two theorems; we put off discussion of one of them (taxes) also to chapter 8.

6.4. Existence and the number of equilibria

We turn next to a brief discussion of two highly mathematical points concerning Walrasian equilibrium. Because the second point especially is very important to one of the central messages of the book, I urge you to pay attention at least to the discussion in regular-sized type.

Existence

So far, we have discussed properties of Walrasian equilibria for exchange economies under the supposition that at least one equilibrium exists. It is, of course, interesting to know whether these results have any relevance: For a given exchange economy, are there Walrasian equilibria?^d In general the answer will be no, so the question really becomes: *Under what conditions on an exchange economy can we guarantee the existence of at least one Walrasian equilibrium?*

To see ways to answer this question, consider an economy with two goods. We continue to maintain all assumptions made earlier: Consumer's preferences are continuous, locally insatiable and nondecreasing. For a two-good economy, this means we can draw some very simple pictures, constructed from the following considerations:

(1) We already know that if (p_1, p_2) is an equilibrium price vector, so is $(\lambda p_1, \lambda p_2)$ for any positive constant λ . Since preferences are nondecreasing, we know that equilibrium prices will be nonnegative, and since equilibrium prices are certainly nonzero, we know that at any equilibrium $p_1 + p_2 > 0$. Putting these facts together, if (p'_1, p'_2) are equilibrium prices, so are $p_1 = p'_1/(p'_1 + p'_2)$ and $p_2 = p'_2/(p'_1 + p'_2)$, and of course $p_1 + p_2 = 1$. In other words, we can restrict attention to prices (p_1, p_2) such that $p_1 + p_2 = 1$.

(2) Fix prices p_1 and $p_2 = 1 - p_1$ such that $p_2 > 0$. Suppose that for a given consumer i , x_1^i is part of a solution to the consumer's utility maximization problem at these prices. Since local insatiability implies that the consumer satisfies her budget constraint with equality at any solution, we know that the corresponding x_2^i must be

$$x_2^i = \frac{p_1 e_1^i + p_2 e_2^i - p_1 x_1^i}{p_2}.$$

(3) Continue to hold fixed a pair of prices p_1 and $p_2 = 1 - p_1 > 0$. Suppose that for each consumer i , x_1^i is part of a solution to consumer i 's utility maximization problem at these prices and suppose that markets clear in good 1 precisely: $\sum_{i=1}^I x_1^i = \sum_{i=1}^I e_1^i$. We assert that this guarantees that we

^d If you pay attention to the more technical material, you know that in the proof of the second theorem of welfare economics we established that Walrasian equilibria exist for at least some initial allocations of the social endowment. For comparison with later discussion in this section, note that this theorem assumed that all consumers have convex preferences.

6.4. Existence and the number of equilibria

have an equilibrium. All that is missing is knowledge that the market in good 2 clears. But

$$\begin{aligned} \sum_{i=1}^I x_2^i &= \sum_{i=1}^I \frac{p_1 e_1^i + p_2 e_2^i - p_1 x_1^i}{p_2} \\ &= \frac{1}{p_2} \left[p_1 \sum_{i=1}^I (e_1^i - x_1^i) + p_2 \sum_{i=1}^I e_2^i \right] = \frac{1}{p_2} \left[p_2 \sum_{i=1}^I e_2^i \right] = \sum_{i=1}^I e_2^i. \end{aligned}$$

The key equality in this string is the next to last, where we use the fact that markets clear in good 1.

We are ready to draw pictures when we make one more assumption:

Assumption 1. For every pair of strictly positive prices p_1 and p_2 , and for each consumer i , the consumer's utility maximization problem has a unique solution.

There are two parts to this assumption. First, this assumption entails the existence of a solution to each consumer's utility maximization problem for every set of strictly positive prices. If you look back to proposition 2.11(a) (and we assume that endowments are nonnegative), you will see that we already have made enough assumptions for this. The second part of the assumption is that the solution is unique. This is more problematic, but proposition 2.11(c) tells us that this is guaranteed if we add strict convexity to our assumptions about the consumer's preferences.

With this assumption, we can write $x_1^i(p_1)$ for the first-commodity part of the unique solution to i 's utility maximization problem at prices p_1 and $p_2 = 1 - p_1$ such that $0 < p_1 < 1$. Note carefully that we only make p_1 an argument of this function, since, by (1), we can restrict attention to prices that sum to one. Also, $x_1^i(p_1)$ explicitly gives demand for the first commodity only; but the demand for the second commodity is implicit, by (2).

Now let $X_1(p_1) = \sum_{i=1}^I x_1^i(p_1)$. This gives the aggregate demand for the first commodity at prices p_1 (and $p_2 = 1 - p_1$). This is what we graph; in figure 6.3, you see the function $p_1 \rightarrow X_1(p_1)$ graphed for arguments p_1 between zero and one. Also in figure 6.3, you see in each picture a horizontal line drawn at the height $e_1 = \sum_{i=1}^I e_1^i$, that is, at the level of the social endowment in good 1. By virtue of (3), at any price $p_1 < 1$ such that $X_1(p_1) = e_1$, we have a Walrasian equilibrium.

Suppose the following two assumptions hold:

Assumption 2. For some (small) value of $p_1 > 0$, $X_1(p_1) \geq e_1$, and for some (large) value of $p_1 < 1$, $X_1(p_1) \leq e_1$.

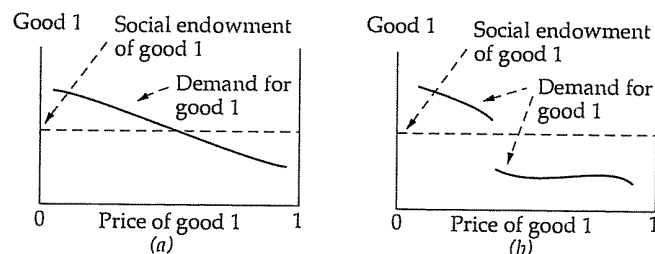


Figure 6.3. Existence of Walrasian equilibrium for the case of two goods.

We have a Walrasian equilibrium at any price p_1 between zero and one (and $p_2 = 1 - p_1$) where $X_1(p_1) = e_1$; that is, where markets clear in good 1 and hence clear in good 2. In (a), we see a case where there is a Walrasian equilibrium, while in (b) we see a case without. Note that, in accordance with assumption 2, we draw in each case aggregate demand that satisfies $X_1(p_1) \geq e_1$ for sufficiently small $p_1 > 0$ and $X_1(p_1) \leq e_1$ for sufficiently large $p_1 < 1$.

Assumption 3. The function $X_1(p_1)$ is continuous in p_1 for $0 < p_1 < 1$.

Under these two assumptions, the intermediate value theorem of calculus would guarantee the existence of a Walrasian equilibrium; a continuous function can't get from a level above e_1 to a level below e_1 without at some point being precisely equal to e_1 .

So are these two assumptions sensible? Begin with the first part of assumption 2. We would like to conclude that for small p_1 the demand for good 1 is no smaller than the social endowment. Consider the indifference curve diagrams in figure 6.4. In (a), we have a consumer with the usual sort of indifference curves. We see there that for low enough prices of the first good — a typical such budget line is sketched in — the consumer will necessarily choose to consume no less than her endowment in the first good. This is clearly the case because all the points that give her at least as much utility as her endowment and that are affordable (the shaded half-lens area) lie to the right of her endowment. If we had this sort of picture for every consumer, then every consumer (for low enough p_1) would demand at least her endowment in the first good, justifying the assumption that $X_1(p_1) \geq e_1$.

But what if the picture for one of the consumers was as in figure 6.4(b). This consumer's indifference curve through her endowment has slope zero at the endowment point. (If you find this an odd case, try drawing the

picture for a consumer whose endowment of the second good is zero.) For this consumer, we wouldn't know that $x^i(p_1) \geq e_1^i$ for all sufficiently small p_1 , and so we wouldn't necessarily have the first part of assumption 2. In essence, the first part of assumption 2 rules out this sort of picture holding for all the consumers; if the price of good 1 gets low enough, someone likes it sufficiently to want (and can afford) enough so that aggregate demand exceeds the social endowment.

For the other half of assumption 2, we consider budget lines that have very steep slope, corresponding to prices p_1 close to one (and hence p_2 close to zero). You may draw the pictures; the basic intuition is the same.

As for assumption 3, if preferences are continuous and demand is single valued (assumption 1), then using techniques similar to the proof of proposition 2.13(b), you can show that demand will be continuous for strictly positive prices.^e So this assumption is entailed by earlier assumptions we made.

To generalize this result, there are essentially three things to think about. First, assumption 1 is too strong. We want to weaken this, but in a way that preserves the "sense" of assumption 3. Second, assumption 2, which can be paraphrased "aggregate demand is well behaved at extreme prices" requires attention. Third, it is clearly desirable to extend beyond the case of only two goods: We'll have to do with something other than the intermediate value theorem.

We will *not* persevere in detail on this subject but instead offer a quick

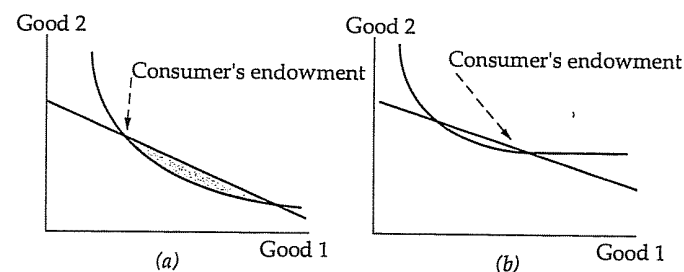


Figure 6.4. Justifying assumption 2.

^e The more mathematically inclined reader can give this a shot, if you didn't already do so when challenged in chapter 2. Or wait a bit for proposition 6.1, where we will generalize this result a step further.

"executive summary." In the references given below and elsewhere, the following happens:

(1) Assumption 1 is simply dropped. Consumers are allowed to have preferences that admit more than one solution at given prices. We redefine $X_1(p_1)$ to be the set of all possible aggregate demands for the first good as a function of p_1 (and, implicitly, $p_2 = 1 - p_1$); that is, $x \in X_1(p_1)$ if $x = \sum_{i=1}^I x_1^i$ where x_1^i is the first component of some solution of i 's utility maximization problem at prices $(p_1, 1 - p_1)$. A function that takes on values that are sets is generally called a *correspondence*; when you hear economic theorists chattering about the *aggregate demand correspondence*, this is what they have in mind.

Not every aggregate demand correspondence will give equilibria. Once we give up uniqueness of the solution to the consumers' problems, we give up automatic continuity, and we need something in place of assumption 3. To see that this is needed, consider first figure 6.5(a). Here we show a consumer's problem where at the prices shown there will be precisely two solutions. It isn't hard to imagine filling in indifference curves so that at other prices, there are unique solutions to the consumer's problem and so that, moreover, demand for good 1 is (except at these relative prices) decreasing in the relative price for good 1. This sort of demand by the individual will lead to aggregate demand as in figure 6.5(b); think of this as aggregate demand for three consumers whose indifference curves look like those in 6.5(a), but with relative prices at which we have two solutions different for the three. With such aggregate demand, there will be no guarantee of an equilibrium, just as depicted in figure 6.5(b).

It turns out that in place of a continuous aggregate demand function we need aggregate demand that is *upper semi-continuous and convex valued*.^f Moreover, we have the following proposition:

Proposition 6.1. *If consumer preferences are continuous, aggregate demand is upper semi-continuous. If consumer preferences are convex, aggregate demand is convex valued.*^g

^f A correspondence $X(p_1)$ is upper semi-continuous if the graph of the correspondence — the set of all points (p_1, x) such that $x \in X(p_1)$ — is a closed set. The term *upper hemi-continuous* is sometimes used instead, reserving *semi-* for functions. And the correspondence is convex valued if for every p_1 the set $X(p_1)$ is convex.

^g Both assertions are for strictly positive prices, since we don't know that aggregate demand is well defined for zero prices. For more mathematically inclined readers, this proposition is a useful result to try to prove.

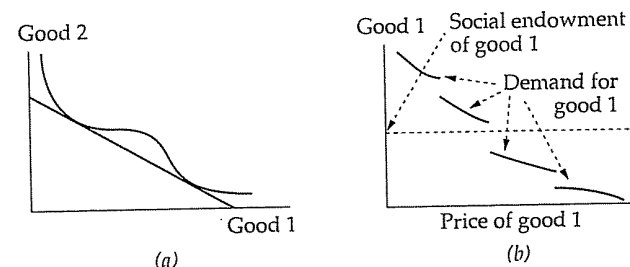


Figure 6.5. Nonexistence of Walrasian equilibrium with demand correspondences.

In (a), we show a consumer whose demand for good 1 takes on two values at a critical price. This can lead to the sorts of discontinuities in aggregate demand shown in (b), which in turn can imply that there is no equilibrium.

So in many general treatments of the existence question, one finds an assumption that preferences are continuous and convex.^h

(2) The problem of well-behaved aggregate demand for extreme prices (or as prices become extreme) is handled in a number of ways. These problems become a bit more involved when there are more than two goods, because then we need to worry about demand when one good has a zero price, when two have zero price, and so on, up to one less than the total number of goods. This matter gets fairly technical fairly quickly, so we leave it for you to read about elsewhere.

(3) When there are more than two commodities, we won't be able to employ the intermediate value theorem so neatly. What then? The mathematical hammer that is used in place of the intermediate value theorem is a *fixed point theorem*. (If you don't persevere through the small type, then at least you should know that when economic theorists are discussing fixed point theorems, the odds are good that they are discussing general equilibrium existence results.)

^h In the higher reaches of mathematical economics, you will find ways even to dispense with convexity of consumer preferences. Without going into detail, the idea is that what we require is aggregate demand that is upper semi-continuous and convex valued. We can get the convex-valued property for aggregate demand even if individual demand is not convex valued, if we have "many" consumers of like characteristics.

There are many fixed point theorems. For our purposes, the two relevant ones are

Brouwer's Fixed Point Theorem. Two things are given: a nonempty, compact, convex set Z (a subset of some Euclidean space, say), and a continuous function $\phi: Z \rightarrow Z$. In this setting the theorem asserts that the function has a fixed point; there is a point $z \in Z$ such that $\phi(z) = z$.

Kakutani's Fixed Point Theorem. The function ϕ in Brouwer's theorem is replaced by a correspondence that associates to each point $z \in Z$ a nonempty subset $\phi(z) \subseteq Z$. This theorem asserts that if ϕ is upper semi-continuous and convex valued, then ϕ has a fixed point — a point z for which $z \in \phi(z)$.

These are fairly heavy mathematical results, but if you know some mathematical programming, some very beautiful algorithmic proofs can be given; see Scarf (1973).

Of what use are these? Suppose there are K commodities. Let Π be the space of all nonnegative price vectors in R^K that sum to one: $p = (p_1, \dots, p_K) \in \Pi$ if $\sum_k p_k = 1$. Let us retreat to the world of assumption 1; the solution of each consumer's problem at any set of prices is unique. In fact, since this is just an "executive briefing," we'll assume this is true for all nonnegative, nonzero prices. (This is clearly a noxious assumption, since one might expect that as the price of some good goes to zero, demand for that good rises to infinity. But this is only an executive briefing.) We can therefore write for each consumer i and commodity k , $x_k^i(p)$ for the demand for good k by consumer i at prices p . Write $X_k(p) = \sum_{i=1}^I x_k^i(p)$ for the aggregate demand for good k .

Assume that $X_k(p)$ is continuous for each k . (Let us say again that this is now for all price vectors, including those where some of the goods have zero price, so this is an overly strong assumption, made only for purposes of illustration.) Consider the following vector function $\phi = (\phi_1, \dots, \phi_K)$ from Π into Π :

$$\phi_k(p) = \frac{p_k + \max(0, X_k(p) - e_k)}{\sum_{k'=1}^K [p_{k'} + \max(0, X_{k'}(p) - e_{k'})]}.$$

Don't panic — this isn't as bad as it looks. Take the numerator first. We add to the old price p_k a positive amount if there is excess demand for good k at prices p . (That makes sense; raise the prices of goods for which there is too much demand.) Then the denominator takes these new relative prices and rescales them so they sum to one again.

This is clearly a function from Π to Π . And we claim that if each $X_k(p)$ is continuous, then so is each ϕ_k , hence so is ϕ . This takes a bit of a proof, but readers with mathematical background shouldn't have a problem with it. Hence by Brouwer's fixed point theorem, this function ϕ has a fixed point.

We claim that if p is a fixed point of ϕ , then p gives Walrasian prices. To see this, note that at a fixed point of ϕ ,

$$p_k = \frac{p_k + \max(0, X_k(p) - e_k)}{\sum_{k'=1}^K [p_{k'} + \max(0, X_{k'}(p) - e_{k'})]}.$$

Note that $\sum_{k'=1}^K p_{k'} = 1$, so if we multiply on both sides of the equation by the denominator, we get

$$p_k \left[1 + \sum_{k'=1}^K \max(0, X_{k'}(p) - e_{k'}) \right] = p_k + \max(0, X_k(p) - e_k).$$

Cancel the common term p_k from both sides, and you get

$$p_k \sum_{k'=1}^K \max(0, X_{k'}(p) - e_{k'}) = \max(0, X_k(p) - e_k),$$

for each k . Multiply both sides of this equation by $X_k(p) - e_k$ and sum over K , and we get

$$\begin{aligned} \left[\sum_{k=1}^K p_k (X_k - e_k) \right] \left[\sum_{k'=1}^K \max(0, X_{k'}(p) - e_{k'}) \right] \\ = \sum_{k=1}^K (X_k - e_k) \max(0, X_k(p) - e_k). \end{aligned}$$

Since Walras' law holds with equality for every consumer, it holds with equality for aggregate demand, which means that the left-hand side of this equation must be zero. Each term in the sum on the right-hand side is nonnegative, so the only way that sum could be zero is if each term is zero, which can only be true if $X_k(p) \leq e_k$ for each k , which is the condition for an equilibrium.

Don't be too taken with this "general" proof. We are back to assuming that the consumers' maximization problems have unique solutions; the proof would be better if we dealt in demand correspondences (which is where Kakutani's fixed point theorem comes in). And, more importantly, our assumption that $X_k(p)$ is well defined and continuous around the edges of Π is really quite outrageous; a better general proof would work quite hard to avoid this.

Nonetheless, the "style" of the proof should be clear. We create a map from the set of prices into itself such that any fixed points are Walrasian equilibria, and then we make assumptions sufficient to guarantee that the map has a fixed point, enlisting a general fixed point theorem along the way.

Finiteness and local uniqueness

So now we know that a Walrasian equilibrium will exist for “well-behaved” exchange economies. We worry next about the other extreme possibility; for a given economy there are so many equilibria that we learn little (except that the equilibrium allocation is Pareto efficient) about what is the outcome of the market process. Have a look at the Edgeworth box in figure 6.6, for example. We’ve drawn things so that three equilibria are shown, and nothing prevents us from continuing to draw indifference curves shaped so that *every* point along the contract curve between the points marked x and x' are Walrasian equilibrium allocations for the initial endowment shown.

But, if we did that, it would only be for this endowment point that we have so many equilibria. For any other endowment point (nearby this endowment point), there would be only a single equilibrium. (Convince yourself of this on the drawing before reading on.) So, in some sense, it would be quite a remarkable “coincidence” if there were so many equilibria for a given exchange economy with a “randomly selected” initial endowment.

This picture is suggestive of some very deep results about the number of equilibria that a “randomly selected” exchange economy may have. One must define what it means for an economy to be randomly selected (and

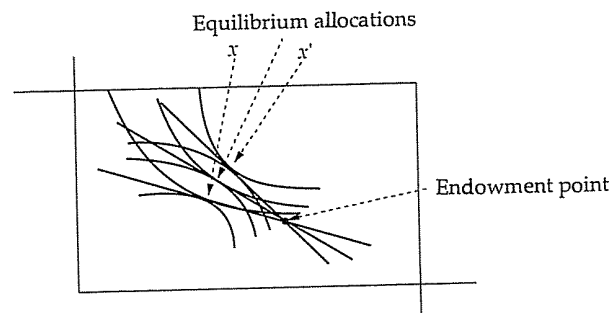


Figure 6.6. Too many equilibria.

From the endowment point marked, three equilibria are shown. We could continue to draw indifference curves so that from this endowment point, *every* point along the contract curve between x and x' is a Walrasian equilibrium allocation. Note that if we do this, then for any other point “close” to the marked endowment point, there is only a single Walrasian equilibrium.

the mathematical methods for doing this are themselves very advanced), but this can be done. And then it is possible to obtain results along the following lines:

Theorem. A “randomly selected” exchange economy will have a finite, and indeed odd, number of equilibria.

To prove this takes methods of differential topology; we won’t even try to explain how this is done here. But we will try to say why this sort of result is of economic (as opposed to mathematical) interest.

Ideally, we would like each exchange economy with which we deal to have a unique Walrasian equilibrium. If this were true, then the theory that a Walrasian equilibrium obtains would make a very definite and precise prediction; we would have enormous scope for testing the theory empirically. We might even be able to go on to economy-wide comparative statics exercises — for example, how does a single consumer’s consumption of a given good change if we increase that consumer’s endowment of the good, or if we increase some other consumer’s endowment of the good? If we could trace how a change in endowment translates into changes in *unique* equilibrium prices and allocations, it would be possible (potentially) to answer this question.

On the other hand, if economies generally have infinite numbers of equilibria, the proposition that an equilibrium will be attained would help us very little. At some given endowment, we might find ourselves at a particular equilibrium. But then if we perturb that endowment, say by giving a fixed consumer more of a particular good, we would have little notion concerning which new equilibrium would pertain. At least, knowing that some equilibrium will pertain would not help us much in predicting how this individual’s consumption would react to this perturbation in her endowment.

The point is that for the sake of definite predictions, having a single equilibrium for every set of initial data is ideal and having an infinite number of equilibria for each set of initial data is terrible.

Unhappily, it is simply unreasonable to suppose that there will be a unique Walrasian equilibrium for a general exchange economy. Conditions on the economy can be given that guarantee uniqueness, but they are very strong.⁷ What the theorem outlined above says is that while the ideal of a unique equilibrium may be unreasonable, we have the “next best” situation of a finite number of equilibria. Moreover, the theorem sketched

⁷ If you go looking for such results, the key words are *gross substitutes*.

can be extended as follows: For most economies, picking any one of its (finitely many) Walrasian equilibria, as we perturb slightly the data of the economy one and only one equilibrium for the perturbed economy will be "close" to the originally selected equilibrium. If we believe that small perturbations of the data of the economy will be met (if possible) by small shifts in the equilibrium, then we can still do local comparative statics for "randomly selected" economies.

To understand this probably takes a more complete discussion than we have given. But an important point is buried here that will return in chapter 14 and at the end of the book in the criticisms that will be given to the application of game theoretic methods. If the preceding discussion is somewhat opaque now, don't worry; you will be returning to this subsection after a while, at which point (I hope and expect) this will make a lot more sense.

6.5. Time, uncertainty, and general equilibrium

As we have already noted, the model of general equilibrium is not clearly relevant to situations where choices will be made through time and under conditions of uncertainty. What does the theory of general equilibrium tell us about choice in a dynamic and uncertain environment?

One approach works within the limited context of choices that are made at any single point in time.

This isn't to say that the future doesn't matter to such choices. The value to the consumer of some of the commodities that are bought or sold at a given point in time is strongly connected to anticipated future prices. For example, preferences over engine size in cars presumably relate to expectations about future gasoline prices. Indeed, some current "commodities" derive value entirely from future considerations. For example, any form of savings or financial investment gives utility only because of the presumed value those savings or investments will have later.

Even so, consumer preferences at a single point in time over current commodities, including financial securities and consumer durables such as cars, can be assumed to conform to the basic models of chapters 2 and 3. If the currently traded commodities have prices, nothing prevents us from analyzing choice in currently open markets according to the methods discussed above.ⁱ

ⁱ There are a few differences: One can borrow money or, in some cases, short sell securities. This would be modeled formally as "consumption" of a negative amount of those

This approach is taken in some of the literature of microeconomics. Work along these lines tends to take a bit more out of the structure of financial assets and the like to put a bit more into the model about preferences for these things. And it tends to work not simply with choice at one point in time, but with choices made serially, where each choice is of the sort analyzed in this chapter, focusing on the connections between choices made at one date and choices made at others. But the basic model of market choice at any single time is just what we have above, where durable and financial assets are treated as commodities. This line of research goes by the general rubric of *temporary general equilibrium*.

But in line with the general tendency to regard dynamic choice as the static choice of an overall strategy, the temporary general equilibrium approach is followed much less frequently than approaches that look at *all* market choices at once.

Arrow-Debreu contingent claims markets

The easiest way to think of looking at all market choices at once is to imagine that all market choices are in fact made at once. A fictional marketplace, known as the market in *Arrow-Debreu contingent claims*, is the central modeling device used here.

We begin by setting the stage. We imagine a finite list of K commodities, each of which can be consumed at any one of a finite number of *dates*, indexed by $t = 0, 1, \dots, T$. Moreover, we imagine that there is uncertainty, which will resolve as time passes. Before consumption at date T takes place, consumers will know which state s from a finite state space S prevails. Between time 0 and time T , uncertainty about the true state s may resolve. This is modeled by a sequence of nondecreasingly finer partitions of S , denoted $F_0, F_1, \dots, F_T$, where $F_0 = \{S\}$ and F_T is the fine partition, where each state is in a set by itself.

For the reader who is unused to the mathematical terminology, a *partition* of S is a set of subsets of S with the properties that each $s \in S$ is in one and only one of the subsets. The members of a given partition are called the *cells* of the partition. A sequence of partitions is nondecreasingly finer if for any two consecutive dates t and $t + 1$, if f is a cell for date $t + 1$ (that is, if $f \in F_{t+1}$), then for some date t cell $f' \in F_t$, $f \subseteq f'$.

The interpretation is as follows. At date t , consumers all know that one of the cells in F_t prevails, and no one knows anything more. An

"commodities." Hence we would have to work in a setting that allowed for negative levels of consumption of some goods. This can pose a number of special problems, especially for questions of the existence of equilibrium.

example may help. Suppose that S has five elements, $\{s_1, \dots, s_5\}$. Suppose that $T = 3$. Then we can imagine that

$$\begin{aligned} F_0 &= \{\{s_1, s_2, s_3, s_4, s_5\}\}; \\ F_1 &= \{\{s_1, s_2\}, \{s_3, s_4, s_5\}\}; \\ F_2 &= \{\{s_1, s_2\}, \{s_3\}, \{s_4, s_5\}\}; \text{ and} \\ F_3 &= \{\{s_1\}, \{s_2\}, \{s_3\}, \{s_4\}, \{s_5\}\}. \end{aligned}$$

This means at date zero, consumers have no idea which state prevails. At date one, they learn that either the state is one of the first two or one of the last three. At date two, if the state is one of the first two, they learn nothing new. And if the state is one of the last three, they learn that either the state is s_3 or one of the last two. Finally, at date three consumers learn whatever is left to learn about which state prevails.

We could depict this example (and any example of this type) with a tree diagram, as in figure 6.7. Time moves from left to right, and at each date, the tree branches reflect information accumulated to that point. So, for example, at the level of the tree corresponding to date 2, the tree has split already into three pieces, one of which (at the next date) branches into s_1 or s_2 , the second of which leads to s_3 , and the third of which branches (at the next date) into s_4 and s_5 .

We stress that we are assuming that at each date t , all consumers have access to precisely the same information. This model does not allow one consumer to have better information about the state of nature than another. Extensions of this basic model do permit consumers to have

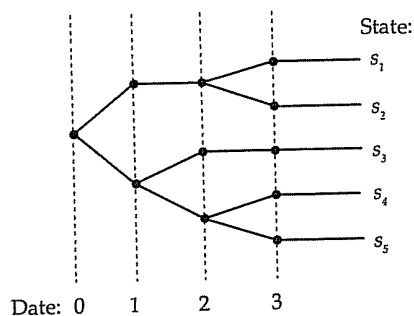


Figure 6.7. Depicting the resolution of uncertainty with a tree.

private information (extensions we won't get to in this book), but the basic model forbids this.

We now add a bit of notation and terminology. For a given date t and a cell $f \in F_t$, we refer to the pair (t, f) as a *date-event* pair. The set of all date-event pairs for date t , which is just $\{t\} \times F_t$ is denoted $\mathcal{E}_t$. And the union over all the dates $t = 1, \dots, T$ of all the date-event pairs is denoted $\mathcal{E}$, which is just $\bigcup_{t=1}^T \mathcal{E}_t$. Don't be put off by the formalisms here; in our example, $\mathcal{E}_0$ has one element, $\mathcal{E}_1$ has two, $\mathcal{E}_2$ has three, and $\mathcal{E}_3$ has five, so $\mathcal{E}$ has eleven elements corresponding to the eleven heavy dots in the tree shown in figure 6.7.

With all this, the consumption space for consumers is the space of all vectors x in the positive orthant X of $(R^K)^\mathcal{E}$. This may look horrible but it isn't. The idea is that the consumer consumes at each date t , and the consumer can change what she consumes according to information she has received at that time. Hence at date 0, she consumes some vector in R^K ; this follows from the fact that F_0 contains a single element. At date 1, she consumes some vector in R^K *contingent upon* which of the two cells in F_1 prevails; this is a vector in the space $(R^K)^{\mathcal{E}_1}$, and so on. A full consumption vector for her is a listing of what she consumes in every date-event pair that she may find herself in, which is $(R^K)^\mathcal{E}$.

For example, suppose in the example there are two goods. Then a consumption bundle for the consumer says how much of each she consumes at date zero, how much she consumes at date 1 if one of the first two states prevails, how much she consumes at date 1 if one of the last three states prevails, and so on. Since there are eleven date-event pairs in the tree in figure 6.7, a consumption bundle for a consumer in this economy would be a point in the positive orthant of R^{22} (or $(R^2)^{11}$).

We assume that each consumer has the usual sort of preferences over X so defined. It is entirely natural to think that since there is uncertainty in this story, we might be able to use the methods of chapter 3 to pin down the nature of the consumer's preferences somewhat. And this is done in most applications. But we will proceed assuming that the consumer has general preferences over X given by a utility function $U_i : X \rightarrow R$.

Finally we come to the most heroic assumption of all. Imagine that prior to date zero consumption there are markets in every good deliverable at every date in every "recognizable" contingency. That is, suppose in our example the two goods are artichokes and broccoli. Then prior to date zero consumption, a consumer can buy or sell artichokes and broccoli for immediate delivery. (The term *spot market* is used for markets in immediately delivered commodities.) But the consumer can buy and sell much more. The consumer can buy and sell, for example, broccoli to be

delivered at date 2 in the event $\{s_1, s_2\}$. This good, which is called an *Arrow-Debreu contingent claim* to broccoli, should be regarded with care. The story is that if you buy ten units of these, when date 2 rolls around, if the state is one of the first two, someone will hand you ten units of broccoli. But if one of the other three states prevails, this claim entitles you to nothing. If you sell one of these claims, you must make it good at date 2 if the state is one of the first two.

Note that we don't imagine markets in date 2—state s_1 broccoli. When date 2 rolls around, consumers are assumed to be unable to tell whether the prevailing state is s_1 or s_2 . Hence we couldn't know whether to make good on a claim against date 2 broccoli contingent on the state being s_1 .

You may recall that way back at the start of choice and demand theory, you were warned that eventually you would be asked to consider commodities of the form — a pint of beer three years from today if the temperature next July 4 is more than ten degrees (Fahrenheit) hotter than the temperature the following June 30. This "commodity" is just a bundle of Arrow-Debreu contingent claims. It's a bundle because it isn't contingent on all the uncertainty that will resolve between now and three years hence. But it is a claim to future consumption contingent on some of the uncertainty that will resolve.

That, essentially, is it. If we imagine that there are, at the outset, markets in every Arrow-Debreu contingent claim, and if preferences are defined on the space X (and endowments lie in the space X), then the theory described early in this chapter can be applied without problem. In earlier developments, we never said what the character of a commodity had to be; so we just apply the theory to the case where commodities include all manner of these contingent claims. Time and uncertainty are fully handled within general equilibrium theory!

Securities and spot markets

Nonsense, you are probably saying. All this is based on the presumption that, before any consumption takes place, there is a complete set of Arrow-Debreu contingent claims markets. Now there are, in the real world, some futures markets where you can contract to receive (or sell) pork bellies or iced broilers at a fixed later date. But you can't buy a pork belly for delivery a year hence contingent on the weather in August. And even the number of futures markets is somewhat limited. There is no futures market whatsoever in broccoli.

General equilibrium theorists have a neat response to this. Suppose that, in our example, the prices of artichokes and broccoli were denominated in dollars. (What's a dollar? Where did this money come from?

These are nontrivial questions that arise in this theory, and I will simply shrug them off here. This isn't to say that these aren't questions that are crucial to this entire story. They are crucial. But this is only going to be an introduction to the subject.) And suppose that, at date zero, there are both spot markets in artichokes and broccoli and markets in claims to \$1 in each date-event pair in the tree. That is, at date zero, for a given price, you can purchase a claim to \$1 in, say, date 2 if the state is either s_1 or s_2 . Note that there are ten date-event pairs other than the date zero pair, so this means twelve markets are open — two spot and ten "financial." Suppose that, moreover, there will be spot markets in the two vegetables at every date, whatever the contingency.

If the assumptions made so far aren't heroic enough, we offer a really big one next: Suppose that, sitting back at time zero, each consumer can precisely and accurately forecast what will be the prices of artichokes and broccoli in their spot markets for each date-event pair. To take an example, suppose consumers know that at date 2 in the event $\{s_1, s_2\}$ the spot price of broccoli will be \$.75 per unit. Suppose as well that at time zero, the price of a claim to \$1 at date 2 contingent on the event $\{s_1, s_2\}$ is \$.40. Then if a consumer wishes to purchase at time zero a unit of date 2—event $\{s_1, s_2\}$ broccoli, she can do so by implementing the following plan: Buy three-quarters of a \$1 date 2—event $\{s_1, s_2\}$ financial contingent claim, which costs \$.30. If at date 2 the event realized is $\{s_1, s_2\}$, this provides her with \$.75. And with this \$.75 she buys her unit of broccoli in the spot market. On the other hand, if she is well endowed with broccoli in this date-event pair and wishes to sell some of it, she can sell date 2—event $\{s_1, s_2\}$ financial contingent claims at date zero, and if at date 2 this event rolls around, she can sell off broccoli to meet her financial obligations.

The point is that if consumers know in advance all the future spot market prices, and if there is a full array of financial contingent claims, consumers can trade from any endowment point $e \in X$ to any desired (and affordable) consumption point $x \in X$ by designing an appropriate dynamic strategy. Money is transferred between date-event pairs using the financial contingent claims, and that money is allocated between consumption goods subsequently in the spot markets. This then implies that all the methods of general equilibrium discussed earlier apply.

Note how strongly we use the assumption that consumers accurately anticipate all future spot market prices. And we assume that consumers follow the standard pattern of chapter 4 in reducing a dynamic choice problem to a static problem of choosing an optimal strategy for dynamic action. In other words, we assume consumers are *hyperrational*, both in their predictive powers and in their planning abilities. These are very

strong assumptions. You will have to judge for yourself whether these assumptions are too strong. (Wait to see applications, which we don't provide here, before making this judgment. These are strong assumptions, but depending on how they influence the conclusions that are drawn in a particular application, you might find them palatable.)

You may still object that even granting hyper-rational agents this isn't a realistic model, because these financial contingent claims don't exist. (Where can you buy a claim to \$1 next August, payable only if the amount of rain in July in New York exceeds the amount of rain in Philadelphia by less than one inch?)

We don't need to have all the financial contingent claims traded at the outset. Suppose, in our example, at date zero a pair of financial contingent claims are traded, one that pays \$1 at date one in the event $\{s_1, s_2\}$ and a second that pays \$1 at date one in the event $\{s_3, s_4, s_5\}$. In date 1—event $\{s_1, s_2\}$, there is a market in a contingent financial claim that pays \$1 at date 2; and in date 1—event $\{s_3, s_4, s_5\}$, there are two financial claims, one that pays \$1 at date 2 in the contingency $\{s_3\}$, and the other paying \$1 at date 2 in the contingency $\{s_4, s_5\}$, and so on. The idea is that at each date in each contingency for that date, there are claims that pay at the next date in every possible following contingency.

Suppose as well that consumers at the outset know what will be the equilibrium prices of these claims as well as the equilibrium prices of spot vegetables. Then a consumer could do just as well as if there were a full array of contingent claims markets. For example, suppose that

At date zero, a claim paying \$1 at date 1 in event $\{s_3, s_4, s_5\}$ costs \$.50.

At date 1 in event $\{s_3, s_4, s_5\}$, a claim paying \$1 at date 2 in event $\{s_4, s_5\}$ will cost \$.60.

At date 2 in event $\{s_4, s_5\}$, a claim paying \$1 at date 3 in event $\{s_4\}$ will cost \$.40.

Broccoli at date zero costs \$2.00 per unit.

Artichokes at date 3 in event $\{s_4\}$ will cost $\$ \frac{2}{3}$ per unit.

Then a consumer, knowing all this, can "trade" date zero broccoli for date 3—event $\{s_4\}$ artichokes as follows: For every unit of date zero broccoli sold (realizing \$2), he buys four claims to date 1—event $\{s_3, s_4, s_5\}$ dollars. At date 1, if the event is $\{s_3, s_4, s_5\}$, he plans to use the \$4 to buy $6\frac{2}{3}$ claims for \$1 at date 2 in event $\{s_4, s_5\}$. At date 2, if the event is $\{s_4, s_5\}$, he plans to use the $\$6\frac{2}{3}$ to buy $16\frac{2}{3}$ claims for \$1 at date 3 in event $\{s_4\}$. And if at date 3 the event is $\{s_4\}$, he plans to use the $\$16\frac{2}{3}$ to buy 25 units of artichoke. That is, the consumer, by buying and selling, can exchange immediate broccoli for date 3—event $\{s_4\}$ artichokes at a ratio of 1:25. If we gave all the prices for all the marketed goods (both vegetable and financial), you (and, presumably, our very smart consumers) could work out the exchange rates for vegetables

in any two date-event pairs. (See problem 9 for a harder example than the one above.)

So all we need are spot markets in vegetables and in financial claims for contingent money at the "next date" and in every possible "next contingency." (We also need very smart consumers.)

You may still be dissatisfied. There are no financial claims around of the type described that pay \$1 in one event and nothing at any other time. There are plenty of financial claims being traded: equity in companies, debt instruments, options on common stock, and so on. But none of it has this simple form. The avid general equilibrium theorist responds to this objection with a still further elaboration: In some (not all) cases, by trading cleverly in the sort of financial instruments just listed, a consumer can manufacture the types of financial claims assumed two paragraphs ago. What is required is that consumers know what prices existing financial instruments will command in every date—event pair, and that there are "enough" of those financial instruments. (The meaning of "enough" is made precise in the literature.)

We will leave you here, with further readings suggested in the bibliographic notes. The basic message to be taken away is that by means of increasingly elaborate constructions, economic theorists extend the reach of the basic model of general equilibrium to cases where time and uncertainty play a part. But often this involves consumers who are both extremely knowledgeable about the future and who are extremely good at planning.

6.6. Bibliographic notes

The classic treatments of general equilibrium theory are Debreu (1959) and Arrow and Hahn (1971). Debreu's monograph gives the essential details, while Arrow and Hahn give a more complete treatment of many of the issues touched on here (and other issues not mentioned here at all). Scarf (1973) presents algorithms for computing Walrasian equilibria, and Scarf and Shoven (1984) show how these computational techniques can be applied. For treatment of issues such as local uniqueness of equilibria for "randomly selected" economies, Mas-Colell (1985), can be consulted by the reader who is very well trained mathematically. (The key tools come from differential topology.) Work on general equilibrium theory is ongoing, for example dealing with economies that have infinitely many commodities and/or infinitely many consumers; a recent "state of the art" treatment of the case of infinitely many commodities is Aliprantis, Brown and Burkinshaw (1989).

Theoretical work on the institutional foundations of general equilibrium belongs for the most part to the journal literature, hence references

given here are apt to be superseded rather quickly. With this in mind, I would recommend Osborne and Rubinstein (forthcoming) for bargaining theoretic foundations.

Experimental work that supports, in limited contexts, the notion of a Walrasian equilibrium is also scattered among the journals. A good starting point is the review article by Plott (1986).

Extending the notions of general equilibrium to models where time and uncertainty play a role takes place in the literatures of both microeconomics and theoretical finance. The notion of a contingent claim is due to Arrow and to Debreu (hence the name Arrow-Debreu contingent claim); some exposition can be found in both Debreu (1959) and Arrow and Hahn (1971). The use of financial claims and "planning" originated in an article by Arrow that is published in English as Arrow (1964). The use of general financial claims and the basic equilibrium concept is formalized in Radner (1972). Since Arrow's and Radner's seminal contributions, there has been a very large literature on the subject; Huang and Litzenberger (1988, chaps. 7, 8) is an accessible introduction. This sort of model becomes especially powerful when trading in securities can take place continuously in time. The theory requires substantial mathematical sophistication, but the reader can get a taste of what goes on in Kreps (1982) and a systematic exposition in Duffie (1988).

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6.7. Problems

- 1. Return to the picture of price equilibrium in the two-consumer, two-good case. For each set of relative prices, we can mark the point that consumer 1 would demand. (Assume that consumers have strictly convex preferences so their maximization problems always have unique solutions.) As we vary the relative prices of the goods, we trace out a curve — the so-called *offer curve* of consumer 1. This is shown for you in figure 6.8.

We can create a similar offer curve for consumer 2, and rotating the picture for consumer 2 by 180 degrees and putting his origin at the location of the social endowment in her (consumer 1's) coordinate system, we get the Edgeworth box with two offer curves. Consider the assertion: *Walrasian equilibrium allocations correspond to points where the two offer curves intersect*. This is not quite correct as it stands. Why not? If you see why not, try to repair the assertion.

- 2. Consider the following exchange economy. There are two goods and two consumers. The two goods are called tillip and quillip and the two

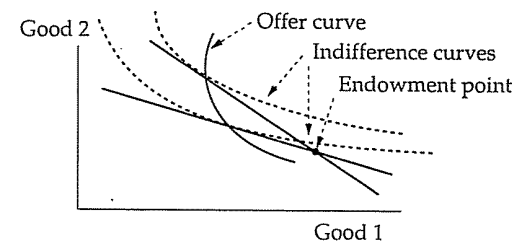


Figure 6.8. The offer curve of a consumer in a two-commodity world.

consumers are called 1 and 2. Consumer 1 has utility function $U_1(t, q) = .4\ln(t) + .6\ln(q)$ (where t is the amount of tillip 1 consumes, and q is the amount of quillip). Consumer 2 has utility function $U_2(t, q) = .5\ln(t) + .5\ln(q)$. Consumer 1 is endowed with 10 units each of quillip and tillip. Consumer 2 is endowed with 10 units of quillip and 5 units of tillip.

(a) What is the Walrasian equilibrium of this economy? (If there is more than one equilibrium, give them all.)

(b) Suppose a social dictator wished to implement an allocation that makes $U_1(t, q) + U_2(t, q)$ as large as possible at the equilibrium. Give all the possible reallocations of the endowment that give the dictator's optimal endowment as a Walrasian equilibrium.

■ 3. A particular social planner I know is very big on mellow consumers. Specifically, she hopes to prevent her consumers from envying each other. To this end, she defines an *envy-free* allocation of resources as one in which no consumer would rather have the consumption bundle assigned to another consumer instead of his or her own. Our social planner wishes to implement an envy-free allocation. She also wishes the allocation to be efficient.

This social planner is also lazy. She isn't willing to figure out the utility functions of her consumers. (She does have a good list of all their endowments.) She is blessed with an economy that functions well as an exchange economy; however she reassigns endowments, the economy finds a Walrasian equilibrium.

Can you help out this social planner? Specifically, describe how to reallocate endowments and shares so that the resulting Walrasian equilibrium is guaranteed to be both efficient and envy free. (Hint: the trick is to find some way to redistribute endowments and shareholdings so that, at every set of prices, consumers all begin with the same level of wealth to spend on consumption. There is a way to redistribute endowments and shareholdings so this is true: What is it?)

■ 4. (a) In the proof of the second theorem of welfare economics, it is claimed that the set

$$X^* = \{x \in X : x \text{ can be allocated among the } I \text{ consumers in a way that strictly Pareto dominates } (\hat{x}^i)\}$$

is convex if preferences are convex. Prove this.

(b) Later in that proof it is asserted that the separating hyperplane vector p is nonnegative. Provide a proof of this assertion.

(c) The remarks after the proof assert that we can replace the assumption that the allocation $(\hat{x}^i)$ is strictly positive with an assumption that each consumer is given a positive amount of at least one desirable good. Show that this assumption will suffice. (Hint: First show that the price of any desirable good must be strictly positive. Although it isn't necessary, to carry out this step you may assume that there is at least one more good and it is in positive supply in the social endowment.)

■ 5. Imagine a three-person economy in which the first commodity is gardening services, the consumption of which makes one's yard more beautiful, and the second good is food. Imagine that two of the consumers in this society live in adjacent houses, while the third lives on the other side of a particularly large mountain. Consumption by the third consumer of gardening services generates no externality for the other consumers, but each of the others generates a positive externality for her neighbor through the consumption of gardening services. To be precise, imagine that consumers 1 and 2 have utility functions of the form

$$V^i(x) = w(x_1^1) + w(x_1^2) + x_2^i$$

where $w : [0, \infty) \rightarrow R$ is a strictly increasing, strictly concave, and differentiable function. Note well that consumers 1 and 2 get just as much utility out of their neighbor's yard as they do out of their own, and their utility for food is linear. (You were warned that this is a very special setting.) Also imagine that consumer 3 has a utility function of the form $V^3(x) = w(x_1^3) + x_2^3$. There is a social endowment of gardening services and food.

(a) Suppose the social endowment is initially allocated evenly among the three consumers. What will be the corresponding Walrasian equilibrium (with externalities)?

(b) Characterize the set of Pareto efficient allocations of the social endowment. Is the equilibrium allocation in (a) Pareto efficient?

■ 6. In a two-person, two-commodity economy, the two commodities are labeled x and y and the two consumers are called A and B. Let x_A , for example, denote the amount of consumption of good x by person A. Each

of the two individuals has initial endowments of one unit of each good apiece. The preferences of a are given by the utility function

$$u_A(x_A, y_A) = \frac{1}{3} \log x_A + \frac{2}{3} \log y_A.$$

The preferences of B are a bit stranger than this. They are given by

$$u_B(x_B, y_B, x_A) = \log x_B + \log y_B + \log(2 - x_A).$$

That is, B gets “disutility” out of A’s consumption of the first good. You may assume throughout that both individuals are locally insatiable (and indeed they are), and you can assume that any solution of first-order conditions and complementary slackness is a global maximum.

(a) What is the Walrasian equilibrium (or equilibria) for this economy? (This is an economy with externalities, so use the definition given in the section on externalities.) You will find that matters are fairly simple if you normalize the prices of the two goods so that they sum to one. Let p be the price of good x so $1 - p$ is the price of good y .

(b) Is the equilibrium allocation in part (a) Pareto optimal? If so, why? (That is, how do you know that it is?) If it is not, how do you know that?

(c) Whether the equilibrium allocation in part (a) is Pareto optimal or not, Mr. B registers the complaint that Ms. A is consuming all too much x for his tastes. Accordingly, a social planner named Lindahl proposes to change the economy as follows. Instead of two prices (for x and y), there will be six. These will be p and $1 - p$ as before, and also q_A , q_B , r_A and r_B . These last four are “transfer prices”; for every unit of x that A consumes, she must transfer q_A units of account (money) to B. For every unit of x that B consumes, he must transfer q_B units of account to A. If A consumes y_A units of good y , she must transfer $r_A y_A$ to B, and r_B gives the transfer price from B to A for B’s consumption of y . The prices q_A , q_B , r_A , and r_B may be positive, zero, or even negative.

Now the economy works as follows. Taking prices p , $1 - p$, q_A , q_B , r_A , and r_B as given (the usual price-taking assumption), A chooses a complete allocation (x_A, y_A, x_B, y_B) for the economy, subject to a single constraint: The net inflow of units of account to A should equal or exceed the net outflow of units from her. That is,

$$px_A + (1 - p)y_A + q_A x_A + r_A y_A \leq p(1 + (1 - p)1 + q_B x_B + r_B y_B).$$

(Recall that A’s endowment is one unit of each of the two goods apiece. This explains the two ones on the right-hand side.) Ms. A chooses the complete allocation to maximize her utility subject only to this constraint; she is under no obligation to make her chosen allocation feasible. Mr. B also chooses a complete allocation, subject to the analogous constraint for him.

An equilibrium for the economy is a set of six prices and a single social allocation such that (1) both consumers faced with those prices independently choose that allocation, and (2) the allocation is socially feasible. In cases where one consumer is indifferent among many different levels in the optimization problem and the second consumer strictly prefers one of those levels, we count this as an agreement in terms of requirement (1).

Find an equilibrium of this economy, for the sort of equilibrium defined above.

(d) Is the equilibrium you computed in part (c) Pareto efficient? If so, how do you know this? If not, how do you know this?

(e) How do the utilities of Ms. A compare in the two equilibria, one from part (a) and one from part (c)? If Ms. A complains to Mr. Lindahl that she, A, is being treated unfairly in this new form of economy, what could Mr. Lindahl do to rectify matters?

(f) If we let Mr. Lindahl loose on the economy of problem 5, what would happen?

■ 7. Parts of this problem may not make sense until chapter 8 is completed. Grapple Synthetics of Boonton (GSB), Inc., has a problem concerning its computer facilities. Its computer facility services 600 identical users with two computers, one of which has a bit more software support than the other, but is more liable to congestion. Specifically, if one of the users uses the first computer and the total number of users of that computer is n , the individual user is able to increase her productivity by an amount $\$(30 - n/10)$. And if one of the users uses the second computer and the total number of users of that computer is m , the user increases her daily productivity by the amount $\$(10 - m/30)$. Assume throughout that these 600 users are all there are; no new ones will appear. The marginal cost of servicing a user is zero. And don’t worry if your answers don’t come out in even integers; fractional users will be okay, for purposes of this problem.

(a) Until now, GSB has not tried at all to regulate use of its computers. Each user has allocated herself to one of the two computers, choosing

so as to maximize her individual productivity gain from the computer under the assumption that other users will stick with the computers they have chosen. Assuming that this is so, what is the equilibrium allocation of users to the two computers? How many are using computer 1; How many are using computer 2; and How many are using neither?

(b) The head of the computer center at GSB is concerned that the current, unregulated method of allocating individuals to computers is not maximizing the total contribution possible for GSB. He suggests the following scheme of transfer prices: A "price per day" will be named for the services of the computers, and users should be told to make use of either computer only if the productivity gain they attain in so doing exceeds this "transfer price." What (single) price per day should GSB name, assuming that its computer users will honestly allocate themselves between computers (or no use) to maximize individual productivity gain less transfer price? What will be the resulting usage for the two computers? (Hint: If you have trouble getting started, try to discover what would happen if the single price of \$5 per day were charged [for each computer].)

(c) The head of the accounting department at GSB thinks that this transfer price scheme is a good idea, but she suggests a more complex scheme in which a different price per day will be charged for each of the two computers. What is the optimal scheme of this sort? What is the resulting pattern of usage for the two computers?

(d) The head of the economic analysis department at GSB is quite confused by the answers you obtained in parts (b) and (c). He says that he recalls from his old days as a microeconomist something called the second theorem of welfare economics, in which it was claimed that efficient allocations of resources should follow the dictates of a (Walrasian) market mechanism. He points out that if the computer services were served by competitive outside vendors (that is, users would actually pay a market price for the services they used), then the competitive price would equal marginal cost, which is zero. Thus zero transfer price should lead to an efficient solution. He suggests that you recheck your algebra in part (b) and (c). In no more than 200 words, explain to this individual why he should, instead, check his economics textbook.

■ 8. With regard to proposition 6.1:

(a) Prove that if consumer preferences are convex, aggregate demand is convex valued. This should be rather easy.

(b) Prove that if consumer preferences are continuous, aggregate demand

is upper semi-continuous. This is done by mimicking the argument made for proposition 2.13(b).

(c) Use (b) to give a fast proof that if consumer preferences are continuous and single-valued, aggregate demand is given by a continuous function.

In all parts of this problem, we are dealing with strictly positive prices only.

■ 9. Consider the set of date-event pairs depicted in figure 6.7. Recall the following data given before:

At date zero, a claim paying \$1 at date 1 in event $\{s_3, s_4, s_5\}$ costs \$.50

At date 1 in event $\{s_3, s_4, s_5\}$, a claim paying \$1 at date 2 in event $\{s_4, s_5\}$ will cost \$.60

At date 2 in event $\{s_4, s_5\}$, a claim paying \$1 at date 3 in event $\{s_4\}$ will cost \$.40

Broccoli at date zero costs \$2.00 per unit

Artichokes at date 3 in event $\{s_4\}$ will cost \$.67 per unit

Add to this the following addition data:

At date zero, a claim paying \$1 at date 1 in event $\{s_1, s_2\}$ costs \$.40

At date 1 in event $\{s_1, s_2\}$, a claim paying \$1 at date 2 in event $\{s_1, s_2\}$ will cost \$.90

At date 2 in event $\{s_1, s_2\}$, a claim paying \$1 at date 3 in event $\{s_1\}$ will cost \$.40

At date 3 in event $\{s_1\}$, artichokes will cost \$1.33

Suppose a consumer wished to sell some date 3-event $\{s_1\}$ artichokes with which she is endowed and use the proceeds to buy date 3-event $\{s_4\}$ artichokes. For every unit of date 3-event $\{s_1\}$ artichokes she sells, how many units of date 3-event $\{s_4\}$ artichokes can she purchase? What is the strategy she follows for affecting such a trade? (This strategy should involve changing her "position" in vegetables at these two date-event pairs only. The key is the first step. She sells date 1-event $\{s_1, s_2\}$ dollars and uses the proceeds from that sale to buy date 1-event $\{s_3, s_4, s_5\}$ dollars.)

chapter seven

The neoclassical firm

In neoclassical economic theory, the firm is an entity, just like the consumer. The consumer has an objective function, utility, that is maximized subject to a budget constraint and any constraints on feasible consumption. The firm has an objective function, profit, that it maximizes subject to constraints imposed by its technological capabilities.

In chapter 2 we spent a lot of time on the consumer's objective function, talking about preferences, choices, and finally utility functions. The budget set constraint was not discussed much. For the neoclassical firm, the reverse is typically true. There is a lot of discussion about how to represent the technological capabilities of the firm and very little about profit maximization.

For the time being, we will keep to this tradition. In this chapter we will describe ways to model the firm's technological capabilities (section 7.1), and then we will discuss the firm's behavior, assuming it chooses inputs and outputs to maximize its profits (sections 7.2 and 7.3). We don't try to explain or justify profit maximization; that discussion is left to much later in the book (chapter 19) on the grounds that we will have little good to say about profit maximization, and only then will we be in a position to give alternatives.

The reader may justifiably wonder, If bad things will be said about profit maximization later in the book, why spend so much time now (and for the next three chapters) using that assumption? There are two reasons: First, this is the traditional model of the firm in microeconomic theory, and more modern developments can best be understood if you understand these classical antecedents. Second, our objections to profit maximization will be on "theoretical grounds" — that is, we don't see a good argument that firms necessarily will or ought to maximize their profits. But that doesn't mean that profit maximization isn't a good positive model. Only the data can speak to that, and then only after we see the implications of profit maximization for observable behavior.

7.1. Models of the firm's technological capabilities

The production possibility set

Firms are presumed to possess productive capabilities. That is, firms are able to transform arrays of commodities into different arrays.

The standard general model is set as usual in a world where there are K commodities. Some of these commodities may be inputs to the firm, some may be outputs of the firm, and some may be either inputs or outputs. Still others may have nothing at all to do with the firm. The firm's productive capabilities are modeled by a set of *netput* vectors in R^K . The term *netput* is used as a generalization of *input* and *output*: For each commodity, we record the firm's production or usage of this commodity, using negative numbers for net inputs and positive numbers for net outputs. For example, suppose there are five commodities: labor (number 1), steel wire (number 2), straight pins (number 3), safety pins (number 4), and wheat (number 5). A typical netput vector for a firm (that makes two types of pins) would be $(-5, -8, 10, 3, 0)$. Interpret this: This firm can take 5 units of labor (say, hours of work), and 8 units (feet?) of steel wire, and make of them 10 units of straight pins and 3 units of safety pins. Another feasible netput vector for this firm might be $(-79, -120, 200, 0, 0)$; with 79 units of labor and 120 units of steel wire, it can make 200 straight pins and no safety pins. Or suppose we had a firm that used inputs one and two to make intermediate product three, and then used more input one and intermediate product three to make product four. In this case, its netput vectors could be either $(-, -, -, +, 0)$ or $(-, -, +, +, 0)$. The first would be appropriate if, in addition to the amount of intermediate good it produced, the firm bought more from the outside. Then it would be a net consumer of the intermediate product, and we would record its net consumption as a negative number. If, however, it sold some of the intermediate good, it would be a net supplier, and the second set of "signs" would be correct. Of course, if it used precisely the amount of the intermediate product that it produced, its netput vector would look like $(-, -, 0, +, 0)$.

The firm's technological capabilities are, then, the set of all the netput vectors of which it is capable. We denote this set of all *feasible* netput vectors for a firm by Z , a subset of R^K . The set Z is called the *production possibility set* or the *technology set* for the firm. Elements $z \in Z$ are called netput vectors or *production plans*.

When $K = 2$, we can draw pictures. Suppose this firm has one input, x (the abscissa component) and one output, y (the ordinate). The picture for Z would be as in figure 7.1.

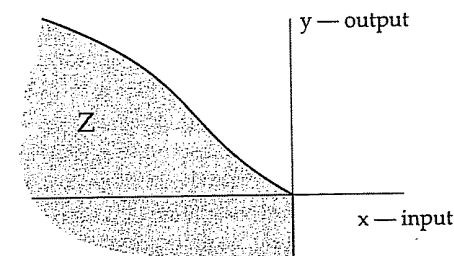


Figure 7.1. A typical production possibilities set.

We sometimes think of the firm's capabilities as varying with the amount of time over which the firm is operating. It is typically assumed that the firm has greater flexibility in what it can do in the long run, so some netput vectors are feasible in the long run but not in the short run. When we wish to make such a distinction, we write Z^l for the firm's long-run production possibility set and Z^s for its short-run production possibilities; the assumption that the firm can do more in the long run than in the short is then written as $Z^s \subseteq Z^l$.

Among the properties that a production set Z might have are:

Convexity: If z and z' are both possible for the firm, then so is $\alpha z + (1 - \alpha)z'$ for each α between zero and one. It is often hard to believe that this assumption is realistic (see nonincreasing returns below), but it is a standard assumption in economics.

Free disposal: If $z \in Z$ and $z' \leq z$, then $z' \in Z$. This is a dumpster theory of production; the firm can always throw out stuff it doesn't want. If one of the outputs of the firm were an effluent, we might doubt this assumption. But as long as more of a commodity is better for someone, better enough so that this someone will come and haul the stuff away, it isn't so bad.

The ability to shut down: $0 \in Z$, where 0 means the commodity vector of all zeros. This seems reasonable if we are thinking of Z as a long-run production possibility set and is less reasonable in the short run. (In the short run, the firm may have contractual obligations to buy inputs, for example.)

Nonincreasing returns to scale: If $z \in Z$ and $0 \leq \alpha \leq 1$, then $\alpha z \in Z$. In words, if U.S. Steel can make steel in a big blast furnace, it can do so as well in scaled-down blast furnaces. This assumption is implied by convexity

and $0 \in Z$, since it consists of taking a convex combination of z and 0 .¹ The name might seem a bit hard to put with the property; it might better be called “nondecreasing returns to scaling down.” (The name will make more sense when we get to production functions.) People will sometimes say decreasing returns to scale and mean nonincreasing returns. Since a minimum efficient scale in blast furnace technology exists (a blast furnace with the capacity for making one pound of steel per day is not likely to be very efficient), this seems a dubious assumption. Nonetheless, it is typically assumed.

Nondecreasing returns to scale: If $z \in Z$ and $\alpha \geq 1$, then $\alpha z \in Z$. Or, if U.S. Steel can run a big blast furnace, they can run two, or three, or five and a quarter of them. This property is sometimes called increasing returns to scale.

Constant returns to scale: If $z \in Z$ and $\alpha \geq 0$, then $\alpha z \in Z$. This is just a conjunction of the previous two properties.

For the one-input, one-output case, the appropriate pictures of increasing, decreasing, and constant returns to scale are easy to draw. If the pictures don't appear instantly in your head, draw them.

Other assumptions are usually added to this list — assumptions of a technical nature that guarantee that the firm's maximization problem (soon to be posed) has a solution. We will wait until such assumptions are necessary before giving the one we will use.

Inputs and outputs: Input requirement sets and iso-quant

In many applications, the commodities that are inputs to the firm are clearly divided from those that are potential outputs. Suppose that we index the K commodities so that commodities 1 through N are inputs, $N+1$ through $N+M$ are outputs, and $N+M+1$ through K have nothing whatsoever to do with the firm. In terms of the firm's production possibility set Z , this division of commodities translates into: If $z = (z_1, \dots, z_K) \in Z$, then $z_k = 0$ for $k = N+M+1, \dots, K$, and $z_k \leq 0$ for $k = 1, \dots, N$. Note that we don't preclude the possibility that the firm has a negative level of its “outputs”; it could buy some of its potential outputs and throw them away or, more realistically, outputs could be potential inputs to later stages of production. This division, then, really binds only on inputs and “no-puts.” We might wish to add one further assumption: If $z_k > 0$ for some $k = N+1, \dots, N+M$, then $z_k < 0$ for at least one k between 1 and N .

¹ But is it equivalent to the two? Draw a picture to see that the answer is no.

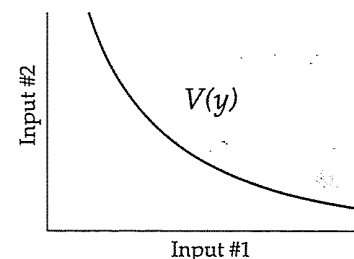


Figure 7.2. A typical input requirement set.

That is, a positive level of any of the outputs requires some nonzero amount of inputs.

When commodities are divided in this fashion into inputs, outputs, and “no-puts,” we will change notation slightly. We'll write $x = (x_1, \dots, x_N)$ for the input vector, where now the levels of the inputs are recorded as positive numbers, and $y = (y_1, \dots, y_M)$ for the levels of the outputs. That is, $(x, y) \in R^{N+M}$ is a feasible input-output combination for the firm if

$$z = (-x_1, \dots, -x_N, y_1, \dots, y_M, 0, \dots, 0) \in Z.$$

For a fixed level $y = (y_1, \dots, y_M)$ of outputs, we can ask if y is a feasible level of outputs for the firm: Is there some feasible netput vector $z \in Z$ that gives this vector of outputs? That is, is there some vector of inputs x such that $z = (-x, y, 0) \in Z$? If so, we can go on to ask which vectors of input x satisfy $(-x, y, 0) \in Z$. Call the set of such $x \in R^N$ the *input requirement set* for making the output vector y and denote it by $V(y)$.²

For the case of two inputs, a typical input requirement set is depicted in figure 7.2. It is typical because it satisfies the following two properties:

Comprehensive Upwards: This is a fancy way to say that if $x \in V(y)$ and $x' \geq x$, then $x' \in V(y)$. This is something like the free disposal assumption on Z , but it is a bit weaker. (Can you draw a picture to show that this is a weaker assumption?)

² Formally, $V(y) = \{x \in R^N : (-x, y, 0) \in Z\}$, with $V(y) = \emptyset$ if it is infeasible for the firm to produce output vector y .

Convex: This is the obvious property. Convexity of each $V(y)$ is implied by, but does not imply, convexity of Z . (Can you provide a proof?)

This all concerns the input requirement set for one vector y of outputs. We may want to discuss the input requirements for several levels at once, in which case it is natural to suppose (and free disposal guarantees) a *nesting* property: If $y \geq y'$, then $V(y) \subseteq V(y')$; i.e., it takes more to make more. (If all input requirement sets are comprehensive upwards, and if they nest in this sense, is free disposal implied?)

When we have this nesting property, the “frontier” of the input requirement set is called the y -output *iso-quant*.^a In intermediate textbooks, production possibility sets are often depicted by the iso-quants, which look rather a lot like indifference curves.

The case of one output: Production functions

We can specialize still further in the case where the firm has a single output, or $M = 1$. (Some of what follows will work for more than one output, but it is rather a mess to draw and explain, so we'll restrict attention to one-output firms.)

In this case, it is natural to suppose that from a given selection of inputs $x = (x_1, \dots, x_N)$ the firm will make as much output y as it can. Hence we have what is called the *production function*, $f(x)$, which gives this greatest amount. Formally, $f(x) = \max\{y : x \in V(y)\}$. Properties of Z translate into properties of f as follows:

Free disposal implies that f is nondecreasing; you can make as much if you have more.

Convexity of Z implies that f is quasi-concave; with a convex combination of inputs, you can make (at least) the lesser of the two levels of output. Alternatively, if each $V(y)$ is convex and if the $V(y)$ nest, then f is quasi-concave. (We'll often go whole hog and assume that f is concave.)

Nondecreasing returns to scale in Z is the same as $f(\alpha x) \geq \alpha f(x)$ for $\alpha > 1$. If an author is fairly careful, the term “increasing” returns to scale will mean that $f(\alpha x) > \alpha f(x)$ for $\alpha > 1$. But some authors will be sloppy and say “increasing” returns to scale when they mean only a weak inequality.

Nonincreasing returns to scale in Z is the same as $f(\alpha x) \leq \alpha f(x)$ for $\alpha > 1$. (Equivalently, we could say $f(\beta x) \geq \beta f(x)$ for $0 \leq \beta < 1$. And remarks

^a Being very formal, this frontier is the set $V(y) \setminus \bigcup_{y' > y} V(y')$.

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similar to those in the preceding property apply to the term “decreasing returns.”)

Constant returns to scale in Z is the same as $f(\alpha x) = \alpha f(x)$, or f is homogeneous of degree 1.

7.2. The profit function

Profits and competitive firms

Having described the firm's technological capabilities, we ask which production plan the firm will adopt. As noted previously, we conform for now to the standard assumption that the firm seeks to maximize its profits.

How do we formulate this objective? Begin with the general model of a firm characterized by the production possibility set Z . We might be very general about things and write $\Pi(z)$ for some function $\Pi : Z \rightarrow R$ as the profits accrued at production plan z . But we can be more specific: Profits are revenues less the costs. If the prices of the K goods are given by the vector $p = (p_1, \dots, p_K)$, then the firm's profits at netput vector z are $\sum_{k=1}^K p_k z_k = p \cdot z$. Note that the sign convention in a netput vector is correct for this formula. Inputs are recorded as negative components in z ; if k is an input, so that $z_k < 0$, the “contribution” to profits $p_k z_k$ is negative (assuming prices are positive).

Writing profits as $\Pi(z) = p \cdot z$ involves an implicit assumption: The firm, by its choice of activities, doesn't change the prices that it faces. We made a similar assumption when we analyzed the consumer, namely that a consumer's choice of consumption bundle didn't change the prices she faced. Our excuse was that, excluding individuals such as J. D. Rockefeller, it is unlikely that any single consumer can change the price of a good by anything worth caring about simply by changing her own demand. Here that excuse doesn't work quite as well. General Motors certainly changes the price of Chevrolets when it changes the number of Chevrolets that it produces. And it probably buys enough steel to have a noticeable effect on the price of steel when it changes its production levels. So for large firms that do affect the prices they face, we might write $p_k(z)$ for the price of commodity k as a function of the production plan the firm selects, and so write the firm's profits as $\Pi(z) = \sum_{k=1}^K p_k(z) z_k = p(z) \cdot z$.

We will rarely be quite this general. We will want, in some places, to consider cases where the firm does change prices of commodities by what it does. But we will almost always be concerned with cases in which the firm's effect on the price of commodity k depends only on the firm's level

of input or output of commodity k . That is, we will suppose that $p_k(z)$ can be written as $p_k(z_k)$, so that $\Pi(z) = \sum_{k=1}^K p_k(z_k)z_k$.

We will use the following terminology. When the firm's level of activities has no effect on the price of commodity k , we will say that the firm is a *price taker* for commodity k , or that the firm is *competitive* in the market for commodity k . When a firm is a price taker in all markets (so that its profits are written very simply as $p \cdot z$ for the market prices p), we will say that the firm is *competitive*.

Pictures from intermediate micro

In general, then, the firm solves the following problem:

$$\text{Maximize } p(z) \cdot z \text{ subject to } z \in Z.$$

We can depict this problem rather easily.

Consider first the case where the firm is competitive, so that the firm's problem simplifies to $\max_{z \in Z} p \cdot z$ for some fixed p . Figure 7.3(a) gives the usual picture for the one-input, one-output case. *Iso-profit* lines are given: lines of the form $p \cdot z$ equal to some constant. Profits increase up and to the right, and the firm picks the point in Z that is furthest up on this scale. This looks quite similar to the consumer's problem except that for the consumer the feasibility set had a simple shape (it was a budget set, with a flat surface), and the iso-objective curves, which are straight lines here, were iso-utility curves, which are typically curved.

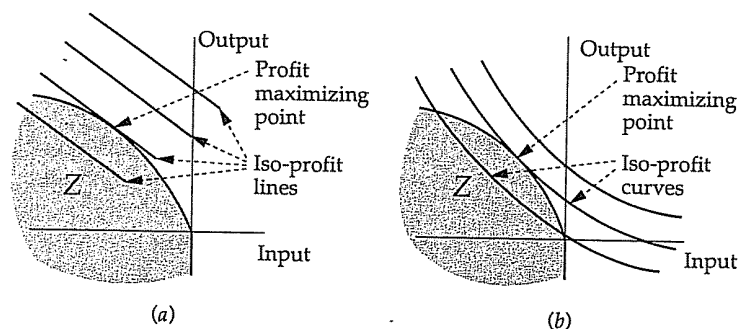


Figure 7.3. The firm's profit-maximization problem for the one-input, one-output case.

In (a), we have a firm with no market power, whereas (b) gives the picture for a firm with market power.

The picture for the firm is a bit more complex if the firm isn't a price taker. Now prices p will depend (by assumption) on the level of the firm's activities z . Iso-profit lines become iso-profit curves — curves along which $p(z) \cdot z$ is constant. In figure 7.3(b) we've drawn the typical picture. Note the convexity of the iso-profit curves. This can be justified by various assumptions about the firm's market power, to which you will not be subjected here. Now the firm looks more complex than the consumer; both the feasible set and the iso-objective curves are curved. But if we considered a consumer who had market power, this consumer would have a convex budget set (under the right assumptions), and the two sorts of problems would be identical mathematically.^b The point is that at this level the mathematics of the firm's and the consumer's problems are extraordinarily similar.

Analysis from intermediate micro

Besides drawing pictures, intermediate microeconomics texts (or, at least, those that use calculus) carry on with some basic analysis of the firm's problem, at least for the case of a firm that produces a single output and so is described by a production function f . Given a production function f and assuming that the firm is competitive, we can write the firm's problem as

$$\max_{x=(x_1, \dots, x_N) \geq 0} pf(x) - \sum_{n=1}^N w_n x_n,$$

where p is the price of the output good and w_n is the price of input factor n . Assuming f is differentiable and the solution is at an interior point,³ the first-order condition for factor n is

$$p \frac{\partial f}{\partial x_n} = w_n.$$

This, in turn, is read according to the verbal formula: *The value of the marginal product of factor n should equal the price of that factor.*

Suppose the firm is competitive in its factor markets but not in its product market. For simplicity, assume that the price of the firm's output

^b Except that in the case of the firm we have to worry whether its set of feasible plans is sufficiently "compact" to guarantee the existence of a solution, something we don't need to worry about in the analysis of the consumer. See also the discussion following the proof of Shephard's lemma.

³ Supply the changes required if one or more of the nonnegativity constraints is binding.

depends only on the level of that output and not (otherwise) on the levels of inputs chosen, so we can write $p(f(x))$ for the price of the output good. The firm's problem becomes

$$\max_{x=(x_1, \dots, x_N) \geq 0} p(f(x))f(x) - \sum_{n=1}^N w_n x_n,$$

which (assuming p is differentiable and we have an interior maximum) gives us the first-order condition (for x_n):

$$[p'(f(x))f(x) + p(f(x))] \left(\frac{\partial f}{\partial x_n} \right) = w_n.$$

This is turned into the verbal formula: *The price of each factor should equal the marginal revenue product of the factor.*

We could go on to cases where the firm has market power in some of the factor markets. If we assume that the price of factor n depends only on the amount of factor n used by the firm, so we can write $w_n(x_n)$, the firm's problem is

$$\max_{x=(x_1, \dots, x_N) \geq 0} p(f(x))f(x) - \sum_{n=1}^N w_n(x_n)x_n,$$

and the first-order condition for factor n (at an interior solution) is

$$[p'(f(x))f(x) + p(f(x))] \left(\frac{\partial f}{\partial x_n} \right) = w'_n(x_n)x_n + w_n(x_n).$$

We can call the right-hand side of this equation the *marginal factor cost* of factor n , and then the first-order condition is verbally rendered: *The marginal factor cost of each factor should equal the marginal revenue product of the factor.*

The profit function for a competitive firm

Suppose we are given a competitive firm, described by a production possibility set Z . At prices (which by assumption do not depend on the z chosen by the firm) given by p , the firm is assumed to pick z to solve the problem

$$\text{maximize } p \cdot z \text{ subject to } z \in Z. \quad \text{FP}(p)$$

7.2. The profit function

When this problem has a solution for given prices p , we write $\pi(p)$ for the value of the objective function. This function π is called the *profit function* of the firm.^c

Does the problem $\text{FP}(p)$ have a solution for every vector of prices p , or even for every strictly positive vector of prices? The answer is no, in general. If, for example, a firm has a constant returns to scale technology, where it can convert n units of a single input (commodity k) into αn units of a single output (commodity k') for any n , and if $p_k < (1/\alpha)p_{k'}$, then our fortunate firm can make infinite profits. Our first order of business, then, is to provide conditions under which we are assured that $\text{FP}(p)$ has a solution.

We make a very strong (and inelegant) assumption. For any two sets A and B in R^K , when we write $A + B$ we mean the set

$$\{x \in R^K : x = a + b \text{ for some } a \in A \text{ and } b \in B\}.$$

We write R_-^K for the negative orthant in R^K ; that is, the set of points $x \in R^K$ all of whose components are nonpositive.

Assumption. There is a compact set Z' such that $Z' \subseteq Z \subseteq Z' + R_-^K$.

Note that if Z itself is compact, it meets this assumption. On the other hand, it is quite possible for Z to have the free disposal property and meet this assumption. But increasing or constant returns to scale (for any nontrivial set Z) would violate the assumption. This is not unexpected; we just saw an example where constant returns led to difficulties.

This assumption gets the result we want, which we state somewhat more generally than we need.

Proposition 7.1. Suppose that Z satisfies the assumption just given, and that $\Pi : Z \rightarrow R$ is a continuous and nondecreasing function. (Nondecreasing means that $\Pi(z) \geq \Pi(z')$ if $z \geq z'$.) Then the problem

$$\text{maximize } \Pi(z) \text{ subject to } z \in Z$$

has a solution, and (moreover) solutions may be found in Z' . In particular, if $\Pi(z) = p \cdot z$ for some nonnegative price vector p , then Π is continuous and nondecreasing, hence $\text{FP}(p)$ has a solution.

The proof is rather easy if you know the relevant mathematics, so easy that we don't even bother to give it as a problem.

This is not the weakest assumption possible to arrive at the desired conclusions. You will find other, weaker assumptions in the literature. Some of

^c Please note that we use Π for profits as a function of the production plan z , whereas π gives maximal profits as a function of (parametric) prices. Do not confuse the two.

the development to follow can be extended to give partial results if no assumption of the form above is made; instead the statements of subsequent results are qualified with clumsy formulae such as: "If $FP(p)$ has a solution..."; or "...for prices where π is defined." You are invited to provide such extensions.

If you skipped the technical excursions, we just guaranteed that $FP(p)$ has a solution for all nonnegative prices p . And with this we can proceed to some results:

Proposition 7.2. Properties of the profit function.

- (a) The profit function π is homogeneous of degree one in prices.
- (b) The profit function is continuous in p .
- (c) The profit function is convex in p .

Part (a) is proved fairly easily. Suppose that z^* solves $FP(p)$. Then at prices λp for λ a positive scalar, z^* gives profits $\lambda p \cdot z^* = \lambda \pi(p)$, hence $\pi(\lambda p) \geq \lambda \pi(p)$. And if $z^\dagger$ solves $FP(\lambda p)$, then at prices p , $z^\dagger$ gives profits $p \cdot z^\dagger$, so that $\pi(p) \geq p \cdot z^\dagger = (1/\lambda) \lambda p \cdot z^\dagger = (1/\lambda) \pi(\lambda p)$, or $\lambda \pi(p) \geq \pi(\lambda p)$. Combining these two inequalities does it.

The proof of part (b) is a little bit technical. Let $p^n \rightarrow p$. Let z^n be a solution of $FP(p^n)$ so that $\pi(p^n) = p^n \cdot z^n$. By proposition 7.1 we can assume that z^n is drawn from the compact set Z' , hence any subsequence of $\{z^n\}$ has a convergent subsubsequence with limit in Z . Look along a subsequence n' in n such that $\lim_{n'} \pi(p^{n'}) = \limsup_n \pi(p^n)$. Then looking along a subsubsequence n'' of the subsequence n' where $z^{n''}$ converges to some $z^0 \in Z$, we have by continuity of the dot product that $\limsup_n \pi(p^n) = \lim_{n''} \pi(p^{n''}) = \lim_{n''} p^{n''} \cdot z^{n''} = p \cdot z^0 \leq \pi(p)$. Conversely, if z solves $FP(p)$, then $\liminf_n \pi(p^n) \geq \lim_n p^n \cdot z = p \cdot z = \pi(p)$. That does it.

The proof of (c) is quite simple. Take any two prices p and p' , and any scalar $\alpha \in [0, 1]$, and let $p'' = \alpha p + (1 - \alpha)p'$. Let $z \in Z$ solve $FP(p'')$, so that $\pi(p'') = p'' \cdot z$. Of course, since z remains feasible at prices p and p' , $\pi(p) \geq p \cdot z$ and $\pi(p') \geq p' \cdot z$. Hence

$$\begin{aligned} \pi(p'') &= p'' \cdot z = (\alpha p + (1 - \alpha)p') \cdot z \\ &= \alpha p \cdot z + (1 - \alpha)p' \cdot z \leq \alpha \pi(p) + (1 - \alpha)\pi(p'). \end{aligned}$$

We go on to properties of solutions of $FP(p)$:

7.2. The profit function

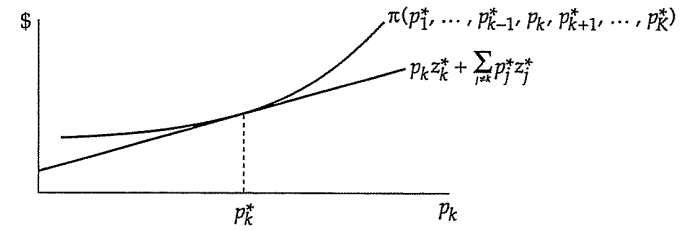


Figure 7.4. Proving Hotelling's lemma.

Proposition 7.3. (a) If z^* is a solution of $FP(p)$, then for any positive scalar λ , z^* is a solution of $FP(\lambda p)$.

(b) If Z is convex, the set of solutions of $FP(p)$ is convex for each p .

Both of these are left for you as easy exercises.^d

Finally, we have a result that has a name. In fact, it has two names:

Proposition 7.4. The Derivative Property, or Hotelling's Lemma. Suppose that the profit function π is continuously differentiable at price vector p^* . Let z^* be any solution of $FP(p^*)$. Then for every $k = 1, \dots, K$,

$$\left. \frac{\partial \pi}{\partial p_k} \right|_{p^*} = z_k^*.$$

If you think about this for a moment, you'll see that this implies that if π is continuously differentiable at p , then $FP(p)$ has a unique solution.^e

The proof can be provided by a simple picture. (If you remember our proof that the derivatives of the expenditure function equal Hicksian demands, you should have no trouble doing this without reading any further.) If z^* is a solution to $FP(p^*)$, then $p^* \cdot z^* = \pi(p^*)$. Since z^* is feasible at any prices p , $p \cdot z^* \leq \pi(p)$. Fix k . In figure 7.4 we graph the functions

$$p_k \rightarrow \pi(p_1^*, \dots, p_{k-1}^*, p_k, p_{k+1}^*, \dots, p_K^*) \text{ and } p_k \rightarrow p_k z_k^* + \sum_{j \neq k} p_j^* z_j^*.$$

^d If you are mathematically inclined, you may wish to continue with more properties of solutions of $FP(p)$. In particular, we will later wonder what ensures that $FP(p)$ has a unique solution. See problems 4 and 5.

^e Math whizzes: Is the converse true?

We have seen that the second of these functions, which is linear, is everywhere below or touching the first, and it touches at $p_k = p_k^*$. By assumption, π is differentiable at p^* , so the derivatives of the two functions must be equal. And the derivative of the linear function is z_k^* .

Or, in words, suppose the price of commodity k changes. We expect that the firm will adjust its production plans. But if the change is small, the difference (beginning from a profit-maximizing position) of staying at the plan z^* or moving to the new optimum is of second order relative to the change in profits overall.

You can also prove this result algebraically, using first-order conditions from the firm's profit-maximization problem.

Of what use is this? Combining this result with the earlier result on the convexity of π gives us some comparative statics results similar to those in chapter 2. To begin, we record a mathematical fact.

Fact. If $\phi : R^K \rightarrow R$ is a convex function and ϕ is twice continuously differentiable at the argument x^* , then the $K \times K$ matrix of second partial derivatives of ϕ , whose (k, j) th element is $[\partial^2 \phi / (\partial x_k \partial x_j)]|_{x^*}$ is positive semi-definite; if $M(x^*)$ is this matrix, then $xM(x^*)x \geq 0$ for all $x \in R^K$. In particular, the diagonal terms of this matrix are nonnegative. Of course, this matrix is also symmetric.

Corollary. Suppose a firm's profit function π is differentiable at prices p^* . Suppose as well that, for all points p in a neighborhood of p^* , $FP(p)$ has a unique solution $z^*(p)$ which, moreover, is continuously differentiable in p . Then the $K \times K$ matrix whose (k, j) th element is $\partial z_j^* / \partial p_k$ is symmetric and positive semi-definite. In particular:

- (a) $\partial z_j^* / \partial p_k = \partial z_k^* / \partial p_j$ for all j and k ; and
- (b) $\partial z_k^* / \partial p_k \geq 0$ for all k .

If you followed developments in chapter 2, you should have no trouble seeing how this follows immediately. Parts (a) and (b) deserve verbal restatements. Part (a) is a bit surprising; it says that the firm's reaction to a penny change in the price of k in terms of netput level z_j equals its reaction in z_k to a penny change in the price of j . You can consider cases where both k and j are inputs, or where both are outputs, but it may be most instructive to think through the case where k is an output and j an input. If the price of output good k rises by a penny, we expect the firm to increase its output and, therefore, to increase the level of its inputs. Increasing the level of input j means that z_j will become more negative, so in this case we expect $\partial z_j^* / \partial p_k < 0$. And if the price of input j rises by

7.2. The profit function

a penny, we expect the firm to cut its output; this is the straightforward $\partial z_k^* / \partial p_j < 0$. Of course, (a) says more than this; it not only says that the signs of the two partial derivatives agree, but also that the two agree precisely in level. (This part is surprising.)

Part (b) is fairly clear. If k is an output and its price rises, we expect the firm will increase its level of that output. Whereas if k is an input and its price rises, we expect that the firm will decrease the amount that it uses of that input, which means that z_k (which is negative) will get closer to zero, which is again a positive partial derivative.

You may rightfully wonder about the status of our assumption that $FP(p)$ has a unique and continuously differentiable solution in a neighborhood of p^* . We will not attempt to justify the differentiability part. But for continuity and uniqueness of the solution, see problems 4 and 5.

Testable restrictions

In our study of the theory of the consumer, we posed the question: What conditions on a finite set of demand data are implied by the model of the consumer as a utility maximizer? We can ask a similar question here. Suppose we see a finite amount of data about the production decisions of a firm. These data are assumed to take the form: At prices p^i , the firm's netput vector is z^i , for $i = 1, \dots, I$. When are these data consistent with our model?

The answer is very simple to give if we maintain the hypothesis that the firm is competitive and that its production possibility set Z hasn't changed over the time frame for which we've gathered these data.^f Suppose $p^j z^i > p^j z^j$ for some $j \neq i$. Then we have a problem. We know that z^i is feasible for the firm. And from the given inequality, we know that, at prices p^j , z^i gives higher profits than z^j . So why was z^j chosen when prices were p^j ?

This consideration gives all the restrictions of the model:

Proposition 7.5. Suppose you are given a finite set of data of the form: At prices p^i , the firm chose netput vector z^i , for $i = 1, \dots, I$. These data are consistent with the profit-maximizing model of a competitive firm (whose production possibility set isn't changing) if and only if for no pair i, j is it the case that $p^j z^i > p^j z^j$.

^f If the firm isn't competitive, the exercise gets quite complicated. Who is setting the prices we observe, and with what objective? After chapters 9 and 10, you could take a run at this sort of analysis with firms that aren't competitive. If we think that the firm's production capabilities may be changing with changing prices or over the time that the data are observed, then there is little we can say. But see problem 6.

The proof is trivial. If there are no obvious violations of profit maximization in the sense of the proposition, then one can take $Z = \{z^1, \dots, z^I\}$. That is, the firm is capable of precisely the I observed netput vectors and no more. (Why does this prove the proposition?) Of course, this is a very odd looking production possibilities set, but we can fill in all the "holes." If we take Z to be the convex hull of the observed netput vectors, or Z to be all netput vectors that are less or equal to points in the convex hull (assuming all prices are nonnegative), we have Z s consistent with the data. In fact, we might ask what is the largest candidate Z consistent with the data? This is easy to give. For each i , let H_i be the half-plane $\{z \in R^K : p^i \cdot z \leq p^i \cdot z^i\}$. Then any Z that contains all the observed z^i and that is contained within $\bigcap_{i=1}^I H_i$ is consistent with the observations.

It may help to give an example for the two-commodity case and draw some pictures. Suppose the first commodity is the firm's input, and the second is its output. Suppose that

- (a) At prices (1, 1), the firm's netput vector is (-3, 5)
- (b) At prices (1, 2), the firm's netput vector is (-6, 7)
- (c) At prices (2, 1), the firm's netput vector is (0, 0)

First check that these data are consistent with the model. At prices (1, 1), the firm chooses a netput vector that yields profits 2; the other two netputs we know to be feasible give (at these prices) profits 1 and 0. So far, so good. At prices (1, 2), the firm chooses (-6, 7), which gives profits 8; the other two netputs we know to be feasible give profits 7 and 0. Okay again. And at prices (2, 1), the firm chooses a production plan that gives profits 0; at these prices the other two known-to-be-feasible production plans give losses of 1 and 5, respectively. So these data are consistent with the model.

Now we draw pictures. Consider the first piece of data. In figure 7.5(a), we put in the point (-3, 5) (as a heavy dot) and through it we drew the iso-profit line corresponding to prices (1, 1). We also put in the two other production plans we saw from the firm as open dots. It is important that they lie below and to the left of the iso-profit line drawn in. We also put in (as dashed lines) the iso-profit lines corresponding to prices (2, 1) and (1, 2) through the production plans selected at those prices. Note once again that in each case the "other" two points lie below and to the left of the iso-profit lines.

What can we say about the production possibility set consistent with these three pieces of data and the model? The set Z must, of course, contain the three production plans that we observed. If Z is convex, it must contain all the points in the heavily shaded region in figure 7.5(b) — the

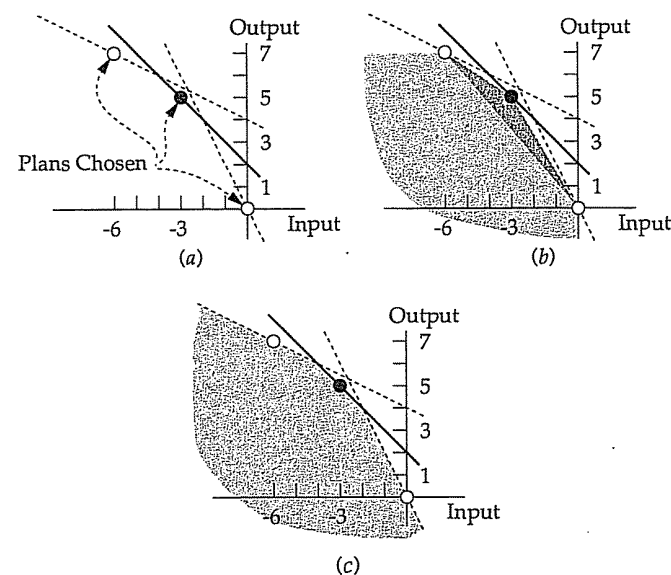


Figure 7.5. Testable restrictions in the model of the firm.

In (a), we show three pieces of data consistent with the standard model of the firm. Each piece of data consists of a production plan and prices at which that plan was chosen. Those prices are used to create iso-profit lines for the plan chosen at those prices. And the data are consistent with the model because all points chosen at some set of prices lie on or below the iso-profit lines so created. In (b), we show the minimal possible Z consistent with the model and these three data points if Z is convex (the heavier shading) and Z is convex and has free disposal (the lighter shading). In (c) we show the largest Z consistent with the model and these three data points: Z must be contained within the intersection of the half-planes lying below and to the left of the iso-profit lines drawn in (a).

convex hull of the three points. If Z is convex and has free disposal, it must contain all the points in the lightly shaded region in figure 7.5(b).

And in figure 7.5(c) we show the greatest possible extent of this production possibility set, consistent with the data and the model. If any point in Z lies above one of the three iso-profit lines through the respective selected plans, that point would have been selected by a profit-maximizing firm instead of the plan supposedly selected. Hence points in Z must lie

below these three iso-profit lines. The shaded region in 7.5(c) is just the intersection of the half-planes referred to previously.

7.3. Conditional factor demands and cost functions

We turn next to the model of the firm with N identified inputs and M outputs. We use the notation developed in section 7.1: The vector x will denote an input vector; y will denote an output vector; $V(y)$ will denote the input requirement set for obtaining outputs at the levels given by y ; and, assuming the firm is competitive in factor markets (which we do throughout this section), $w = (w_1, \dots, w_N) \in R^N$ will denote the prices of the various inputs or factors of production.

Suppose the firm, for whatever reason, fixes on the idea that it must produce outputs at levels y . It still will pick inputs in order to minimize its costs of production; that is, it will solve

$$\text{minimize } w \cdot x \text{ subject to } x \in V(y). \quad \text{FCMP}(w, y)$$

Here, FCMP are the initials of the firm's cost-minimization problem.

We have fewer worries about the problem than we do about $\text{FP}(p)$. In particular, existence of a solution is much less a problem. Assuming $V(y)$ is nonempty and closed, if the factor prices w are strictly positive, then we are guaranteed a solution: Take any point $x \in V(y)$. Since x is one feasible way to produce y , costing $w \cdot x$, the optimal solution costs no more. That is, the optimal solution must come from the set $\{x' \in V(y) : w \cdot x' \leq w \cdot x; x' \geq 0\}$. If $V(y)$ is closed, this set is compact, ensuring the existence of a solution.

We write $c(w, y)$ for the value of this cost-minimization problem, as a function of the two sets of parameters, prices w and output level desired y . This function c is called the *cost function*.

For the case of two inputs, we can give a very revealing depiction of $\text{FCMP}(w, y)$. (If you have taken a course in intermediate micro, you are sure to have seen this picture.) In figure 7.6, *iso-cost lines* are parallel straight lines, and the firm will select that point in $V(y)$ that is as far down in terms of cost as possible. This picture should remind you of figure 2.4, the picture for the Dual Consumption Problem that defines the expenditure function and Hicksian demand. Instead of a production isoquant for a given level of production, we had there an indifference curve for a given level of utility. Instead of minimizing cost, we were minimizing

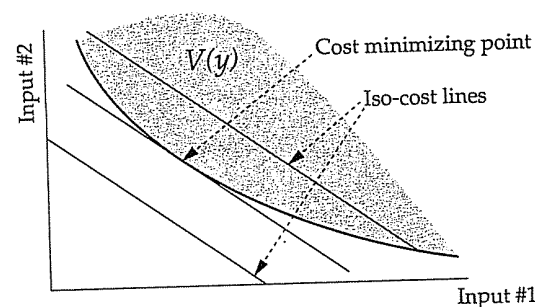


Figure 7.6. Finding the cost function in the two-input case.

expenditure. But the pictures are the same, and so, it turns out, are the mathematics.

Proposition 7.6. Properties of the cost function.

- (a) The cost function is homogeneous of degree one in w for each fixed y .
- (b) The cost function is nondecreasing in w for each fixed y .
- (c) The cost function is concave in w for each fixed y .

Parts (a) and (b) are left for you to prove. The proof of (c) should be old hat by now, but let us give it a final time. Choose any two factor price vectors w and w' and some $\alpha \in [0, 1]$, and let $w'' = \alpha w + (1 - \alpha)w'$. Let x be any solution of $\text{FCMP}(w'', y)$. Then $x \in V(y)$, and x could be used to produce y at prices w and at prices w' . Of course, at these two sets of factor prices something might be better than x , so that $c(w, y) \leq w \cdot x$ and $c(w', y) \leq w' \cdot x$. And thus $\alpha c(w, y) + (1 - \alpha)c(w', y) \leq \alpha w \cdot x + (1 - \alpha)w' \cdot x = w'' \cdot x = c(w'', y)$.

Solutions of $\text{FCMP}(w, y)$ are called *conditional factor demands*, where "conditional" refers to the fact that the output vector y is fixed.

Proposition 7.7. Properties of solutions of $\text{FCMP}(w, y)$.

- (a) If x solves $\text{FCMP}(w, y)$, then x solves $\text{FCMP}(\lambda w, y)$ for all strictly positive scalars λ .
- (b) If $V(y)$ is convex, then the set of solutions of $\text{FCMP}(w, y)$ is convex.⁹

⁹ See problems 4 and 5 for more and, in particular, for conditions sufficient to guarantee

The proofs are left to you. You should have no problems by now.

Proposition 7.8. Shephard's Lemma. Suppose that $c(w, y)$ is continuously differentiable in w (for fixed y) at price vector w^* . Let x^* be any solution of $\text{FCMP}(w^*, y)$. Then

$$\left. \frac{\partial c(w, y)}{\partial w_n} \right|_{(w^*, y)} = x_n^*.$$

As noted immediately following Hotelling's lemma, this implies that if $c(w, y)$ is continuously differentiable in w at w^* , then $\text{FCMP}(w^*, y)$ necessarily has a unique solution.^h

This should be compared to proposition 2.15. It is essentially the same result, although we're a bit fancier here by noting that $\text{FCMP}(w^*, y)$ must have a unique solution if c is continuously differentiable in w rather than assuming uniqueness of the solution. In any event, the proof is precisely the same; use the same picture as figure 2.6, although the names of the various objects change.

Time for a really technical aside. We have four applications of the "envelope theorem," namely proposition 2.15, Roy's identity (proposition 2.16), Hotelling's lemma (proposition 7.4), and now Shephard's lemma. Of the four, all but Roy's identity were extremely easy, whereas Roy's identity was only easy. A difference between Roy's identity and the other three explains why the three were easier. In proposition 2.15, Hotelling's lemma, and Shephard's lemma, prices, as they change, change the objective function in the optimization problem being studied. But the feasible set stays the same. In Roy's identity, as prices change, the objective function (utility) stays and the feasible set (the consumer's budget set) changes. The latter is simply a bit harder. You can't just sit at the old optimal solution and see what its value is at the new prices; to remain feasible you have to move a bit from the old optimal solution. If you can make sense of all this, think through: What would be the analogous result for the following (quite daft) problem for a firm with one output? Given a fixed budget to be spent on inputs, maximize the amount of output you can produce, as parameterized by the prices of the inputs.

Suppose that for fixed output y , $\text{FCMP}(w, y)$ has a unique solution for each w in some neighborhood, and suppose, moreover, that these solutions, viewed as a function $w \rightarrow x_n^*(w, y)$, are continuously differentiable. By virtue of proposition 7.6(c) and 7.8, we can now start developing results of the usual sort based on the observation that the $N \times N$ matrix whose

that $\text{FCMP}(w, y)$ has a unique solution.

^h Same challenge as before: What is the status of the converse?

(ℓ, n) th term is $\partial x_\ell^* / \partial w_n$ is a symmetric, negative semi-definite matrix. You ought to be capable of discerning the implications of this by now.

7.4. From profit or cost functions to technology sets

(This entire section is in small type for the technical reader.) In chapter 2, we posed the question: Given a set of alleged Marshallian demand functions, are the given functions in fact demand functions for some consumer conforming to the standard model? The reason for asking this question, we said, was to build tractable models. It is hard to get closed form demand functions from utility functions that give the sorts of parametric flexibility we might like. But if we have an answer to the question, then we can start with demand functions that have the desired parametric features and check that they are in keeping with the standard model.

For the same reason, we wish to be able to answer the question: Given a profit function π , or given a cost function c , under what conditions are we certain that these arise from profit-maximizing behavior by a competitive firm? We are interested in these questions because it might be convenient to start with a profit function or a cost function instead of with a production possibility set or a collection of input requirement sets.

We could, as well, put the question: Are given conditional demand functions x^* (whose arguments are pairs (w, y)) consistent with profit maximization by a competitive firm? And although we didn't introduce the terminology previously, we can ask: Suppose we are given supply/demand functions z^* (with domain the space of prices R_+^K and range the space of netputs R^K) that are alleged solutions of $\text{FP}(p)$ for some competitive firm. Are they in fact consistent with the standard model?

We will give answers here for the profit and supply/demand functions. We will assume that these given functions are smooth; the profit function (if that is what we are given) is twice continuously differentiable, and the supply/demand functions (if that is what we are given) are continuously differentiable. You may wish to adapt our results and analysis to cost functions and conditional factor demand functions and (what is harder) to less smooth candidate profit and supply/demand functions.

Proposition 7.9. (a) If a candidate profit function π is homogeneous of degree one and convex, and if we define supply/demand functions z^* from π by $z_k^*(p) = \partial \pi / \partial p_k$, then these supply/demand functions are homogeneous of degree zero, and the $K \times K$ matrix of terms $\partial z_j^* / \partial p_k$ for $j, k = 1, \dots, K$ is symmetric and positive semi-definite.

(b) If candidate supply/demand functions z^* are homogeneous of degree zero, and the $K \times K$ matrix of terms $\partial z_j^* / \partial p_k$ for $j, k = 1, \dots, K$ is symmetric and positive semi-definite, then the profit function π defined by $\pi(p) = p \cdot z^*(p)$ is homogeneous of degree one and convex.

(c) In either case (a) or (b), if we define from π the set Z given by

$$Z = \{z \in R^K : p \cdot z \leq \pi(p) \text{ for all } p \geq 0\},$$

then π and z^* correspond to a competitive firm that has production possibility set Z .

To prove this we need to know:

- (1) Euler's law: If $\phi : R^K \rightarrow R$ is a differentiable function that is homogeneous of degree one, then $\phi(x) = \sum_{k=1}^K x_k (\partial \phi / \partial x_k)$.
- (2) If ϕ is homogeneous of degree one, the functions $\partial \phi / \partial x_k$ are all homogeneous of degree zero.
- (3) If ϕ is homogeneous of degree zero, $\sum_{k=1}^K x_k (\partial \phi / \partial x_k) = 0$.

These are easy to prove: Differentiate the identity $\phi(\lambda x) = \lambda \phi(x)$ first with respect to λ for (1) and then with respect to x_k for (2). And for (3), differentiate the identity $\phi(\lambda x) = \phi(x)$ with respect to λ .

Thus if π is homogeneous of degree one, the supply/demand functions defined in (a) are homogeneous of degree zero. And the convexity of ϕ gives the corresponding properties on the matrix of mixed partials of z^* . This is (a).

Conversely, if we have supply/demand functions that are homogeneous of degree zero, defining π from them as in (b) clearly gives us a function homogeneous of degree one. For this π ,

$$\begin{aligned} \left. \frac{\partial \pi}{\partial p_k} \right|_p &= z_k^*(p) + \sum_{k'=1}^K p_{k'} \frac{\partial z_{k'}^*}{\partial p_k} \\ &= z_k^*(p) + \sum_{k'=1}^K p_{k'} \frac{\partial z_k^*}{\partial p_{k'}} = z_k^*(p), \end{aligned}$$

where the next to last equality follows from the presumed symmetry, and the last equality follows from (3). And then the positive semi-definiteness of the matrix of mixed partials of z^* implies that π is convex. This gives (b).

Finally, for (c), we need to show that for every p , $z^*(p)$ is the solution to the profit-maximization problem at prices p where choice is out of Z as defined. It is immediately clear that no alternative production plan from Z will give higher profits than $z^*(p)$, since Z is defined precisely so this is so. The key is to show that $z^*(p)$ is in Z as defined, which means that $p' \cdot z^*(p) \leq \pi(p')$ for all $p' \geq 0$. This can be shown as follows: Since π is convex, we can show that

$$\pi(p') \geq (p' - p) \cdot D\pi(p) + \pi(p) = p' \cdot D\pi(p) - p \cdot D\pi(p) + \pi(p),$$

where $D\pi(p) = (\partial \pi / \partial p_1, \dots, \partial \pi / \partial p_K)|_p$. (Consult a good book about convex functions for this result.) But by (1), $p \cdot D\pi(p) = \pi(p)$, so this inequality simplifies to $\pi(p') \geq p' \cdot D\pi(p)$. And $D\pi(p) = z^*(p)$, so the inequality is $\pi(p') \geq p' \cdot z^*(p)$, which is just what we want.

Besides giving us useful test conditions for when a profit function or set of supply/demand functions corresponds to a competitive profit-maximizing

firm, this result, or rather its proof, illustrates how one recovers the "relevant technology" of a firm from profit functions and/or supply/demand functions. The definition of Z given in the proposition is the key. Recall that in the subsection on testable restrictions, we said that the largest possible production possibility set consistent with a given set of data was the intersection of half-planes given by those data. If you look at the definition of Z that is given, you will see that it is nothing more than the intersection of the infinitely many half-planes given to us by the supply/demand function! And then the real magic in the proof is the demonstration that $z^*(p)$ for each p is in that intersection, which is where the convexity of π comes in.

7.5. Cost functions and -runs

In section 7.3, we investigated properties of the cost function in factor prices for fixed output level y . Now we suppose that w is fixed (as a function of x) and look at the cost function as a function of output levels y . We will deal throughout with the case of a one-output firm.ⁱ

Assume we have a one-output firm, and assume the technology of the firm is described by the production function $f(x)$. Fix factor prices w , which we will allow to depend on the levels of the factor inputs. Define, for each production level y ,

$$TC(y) = \min\{w(x) \cdot x : y = f(x)\}.$$

That is, $TC(y)$ gives the cost of the least-cost combination of factor inputs from which y can be produced. TC stands for *total cost*; we append the T because we will shortly deal with other modifiers (average, marginal). Note that we do not require that the firm is competitive in factor markets; compare with our treatment of cost functions in section 7.3.

Assuming we have the cost function TC , the firm's problem is to maximize $p(y)y - TC(y)$. The first-order condition is $p'(y)y + p(y) = TC'(y)$, or *marginal revenue equals marginal cost*. The term $\epsilon(y) = p(y)/yp'(y)$ is called (for various reasons) the elasticity of demand facing the firm; with this definition, we can rewrite this first-order condition as

$$p(y) \left(1 + \frac{1}{\epsilon(y)} \right) = TC'(y).$$

ⁱ Everything that follows concerning total and marginal costs would work easily for multiproduct firms. But what we will say about average costs doesn't extend at all easily; if you have ever been involved in the allocation of joint costs among several products, you will see why immediately.

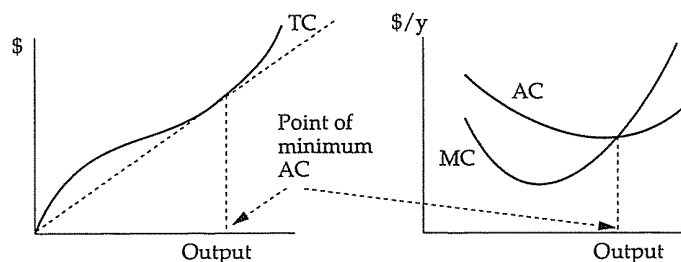


Figure 7.7. Total cost, average cost, and marginal cost curves.

If the firm has no market power in the product market, so that $p'(y) = 0$ and $1/\epsilon(y) = 0$ (the firm faces a perfectly elastic demand curve), the first-order condition simplifies to $p(y) = TC'(y)$, or *price equals marginal cost*.⁴

As for the geometry of the cost function, it is typical to define the average cost function by $AC(y) = TC(y)/y$ and marginal cost by $MC(y) = TC'(y)$, obtaining the picture in figure 7.7. Average costs always will be falling when marginal cost is less than average cost, and average costs are rising when marginal cost is greater than average cost: To see this differentiate $AC(y)$ to get $AC'(y) = TC'(y)/y - TC(y)/y^2 = (1/y)[MC(y) - AC(y)]$. Hence whenever $MC(y) = AC(y)$, we are at a local minimum or maximum (or, if we want to list all the cases, some other critical point) of AC .

Suppose our technology exhibits increasing returns to scale over some region. Then AC will be falling (or, at least, not rising), hence $MC \leq AC$. This takes a proof, which you are asked to supply in problem 10. You are also asked in problem 10 to show that the converse is not necessarily correct. But when there are constant returns to scale, AC will be flat, which means that $AC = MC$.

Fixed and variable costs as long- and short-run costs

In a number of models, the distinction between what a firm can do in the long and short runs is modeled by supposing that there are certain inputs whose levels cannot be shifted at all in the short run but that are freely variable in the long run. In terms of the general formulation in which a firm's technology is described by Z , we would begin with a

⁴ If the "equation" $1/\epsilon = 0$ when $\epsilon = k/0$ bothers you, just use the first-order condition $p'(y)y + p(y) = TC'(y)$.

long-run production possibility set Z^l ; there would be commodities, say $k = 1, \dots, K'$, which are fixed in the short run, and for levels $(\hat{z}_1, \dots, \hat{z}_{K'})$ of those short-run fixed levels of netput, the short-run production possibility set of the firm would be

$$Z^s = \{z \in Z^l : z_k = \hat{z}_k \text{ for } k = 1, \dots, K'\}.$$

Note that we should index this set Z^s by $(\hat{z}_1, \dots, \hat{z}_{K'})$; as those levels change, so does Z^s . Note also that this accommodates short-run fixed levels of factor inputs and short-run fixed levels of required outputs. This isn't entirely general: If the firm has short-run commitments that it must meet but can exceed (say, it must supply $\hat{z}_K$ of good K), then it would make more sense to define Z^s where the restriction on the level of z_K is $z_K \geq \hat{z}_K$.

Now consider the special case of a one-output firm whose technology is given by a production function formulation and that is competitive in its factor markets. The short-run restrictions are that levels of some of the factor inputs are set. As usual, we index factor inputs by $1, 2, \dots, N$, and we let $1, \dots, N'$ index those that are fixed in the short run, and $N'+1, \dots, N$ index those that are freely variable in the short run. In the short run, then, we are stuck at some level of factor inputs $(\hat{x}_1, \dots, \hat{x}_{N'})$, paying a bill $\sum_{n=1}^{N'} w_n \hat{x}_n$ for them.⁵ If we wish to produce y units of output, we minimize short-run costs by solving

$$\min_{x_{N'+1}, \dots, x_N} \left\{ \sum_{n=N'+1}^N w_n x_n : y = f(\hat{x}_1, \dots, \hat{x}_{N'}, x_{N'+1}, \dots, x_N) \right\}. \quad (*)$$

The cost of the fixed factors, $\sum_{n=1}^{N'} w_n \hat{x}_n$, is called the *fixed cost*, or FC . The minimum cost of the variable factors, given by $(*)$, is called the (short-run, total) *variable costs*, or $TVC(y)$. (Implicitly, this function depends on the levels of the fixed factors.) Of course, only TVC depends on y . Finally, total costs are $TC(y) = FC + TVC(y)$.

At this point, we can define average variable costs, average fixed costs, average fixed-plus-variable costs, and marginal costs. The pictures are as in figure 7.8. Since the variable costs, average variable costs, average total

⁵ For the remainder of this section, we will assume that the firm is a price taker in its factor markets. You might ponder what changes if (1) w_n is a function of x_n or (2) w_n is a function of the entire vector x . In (2), consider in particular what happens if w_n for some $n \leq N'$ depends on $x_{n'}$ for some $n' > N'$.

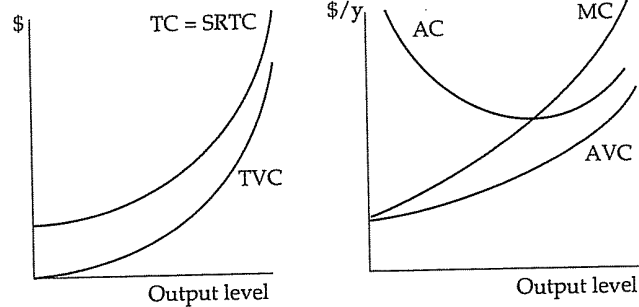


Figure 7.8. Short-run cost curves.

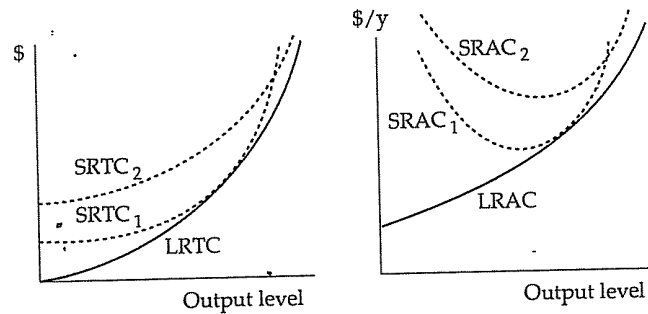


Figure 7.9. Short- and long-run cost curves.

(fixed-plus-variable) costs, and marginal costs all are based on the short-run consideration that some of the factors can't be varied, they are also sometimes called short-run variable, short-run average variable, short-run average fixed-plus-variable, and short-run marginal costs.

And, to conclude, we can in the long run vary all the factor inputs to give us long-run total, average, and marginal costs. As we change the quantities of the fixed factors, we trace out all the possible short-run cost curves. Long-run cost curves (of the total and average variety) are the lower envelopes of the various short-run cost curves. See figure 7.9. Note that, in figure 7.9, some short-run average cost curves lie everywhere above the long-run average cost curve. A SRAC curve is tangent to the LRAC curve only if the fixed amounts of the short-run fixed factors are optimum overall for some level of output. If (because, e.g., factor prices changed unexpectedly) the firm is "stuck" with fixed factor amounts that are not optimal for any long-run level of production, then the firm will have SRTC and SRAC curves that are everywhere strictly above the LRTC and LRAC curves.

7.6. Bibliographic notes

Varian (1984) covers a number of topics not covered here, including remarks on the geometry of cost functions, conditional factor demand functions, and input requirement sets. (If an iso-quant is flat, what does that mean for conditional factor demands? If an iso-quant is kinked, what are the implications?) He also introduces the problems of empirical estimation of production and cost functions.

We have been very loose in our treatment of technical issues such as differentiability of demand/supply and conditional factor demand functions. All this can be done rigorously; see Sonnenschein (1987) for the gory details.

References

- Sonnenschein, H. 1987. *Lecture Notes in Microeconomic Theory*. Princeton University. Mimeo.
- Varian, H. 1984. *Microeconomic Analysis*, 2d ed. New York: W. W. Norton.

7.7. Problems

1. Below are six production functions for a firm that turns two variable factors of production, capital k and labor l , into a single output y . Labor is freely variable in the short and the long run. Capital is fixed in the short run, but can be varied in the long run. Assume that the firm is a price taker in the factor markets, that the price of capital is r , and the price of labor is w . For each of these six production functions, what is the firm's long-run total (variable) cost function? What is the firm's short-run total (variable) cost function (given a level k of capital)? In the short and in the long run, does the firm exhibit increasing, decreasing, or constant returns to scale?

- $f(k, l) = k^\alpha l^\beta$ for $\alpha > 0, \beta > 0$, and $\alpha + \beta < 1$.
- Same as part (a), but $\alpha + \beta = 1$.
- How about $\alpha + \beta > 1$?
- $f(k, l) = (\min[k/\alpha, l/\beta])^\gamma$, where $\alpha > 0, \beta > 0$, and $0 < \gamma < 1$.
- Same as (d), but $\gamma = 1$.
- $f(k, l) = \alpha k + (1 - \alpha)l$.

■ 2. (a) Show that if, for a given firm, Z has free disposal, then $V(y)$ is comprehensive upwards for every y . (Assume the firm has N inputs and M outputs.) Show by example that the converse is false. If each $V(y)$ is comprehensive upwards and the nesting property holds (if $y \geq y'$, then $V(y) \subseteq V(y')$), does the corresponding set Z have free disposal?

(b) Show that if Z is convex, then each $V(y)$ is convex. Show by example that the converse is false.

■ 3. Suppose a one-output firm is described by a production function $f(x)$. Show that if the firm has free disposal, then f is nondecreasing. Show that convexity of Z implies that f is quasi-concave. Show that nondecreasing returns to scale in Z implies $f(\alpha x) \geq \alpha f(x)$ for $\alpha > 1$. Is the converse true?

■ 4. Let $Z^*(p)$ be the set of all solutions of $FP(p)$ for a firm whose production possibility set Z is closed. Show that the correspondence $p \rightarrow Z^*(p)$ is upper semi-continuous. Show that if $FP(p)$ has a unique solution for all prices in some region, then the function $z^*(p)$ defined over that region is continuous. State and prove parallel results for the correspondence $p \rightarrow X^*(p)$, of solutions of $FCMP(w, y)$.

■ 5. For a given production possibility set Z , we say that $z \in Z$ is *efficient* if there is no $z' \in Z$ such that $z' \geq z$ and $z' \neq z$. Suppose that Z is convex and has the property that if z and z' are both in Z and $\alpha \in (0, 1)$, then $\alpha z + (1 - \alpha)z'$ is not efficient. Show that this implies that, for a competitive firm facing strictly positive prices p , the solution to $FP(p)$ is unique. Construct a parallel definition for input requirement sets $V(y)$ and give a parallel result concerning the uniqueness of solutions for $FCMP(w, y)$.

■ 6. Suppose that we have obtained a sequence of observations of the production plans adopted by a firm for a sequence of prices. Specifically, we observe at one time that, at prices p^1 , the firm chooses production plan z^1 ; at a later time, at prices p^2 , the firm chooses plan z^2 ; and so on, with the last observation being that at prices p^I the firm chooses plan z^I . The model we have is that the firm is competitive and chooses a profit-maximizing plan from its production possibility set at each point in time, but that the production possibility set may, because of technological advances, be growing through time. That is, at the time of the first choice, the firm has production possibility set Z^1 ; at the second time, the firm has a production possibility set Z^2 with $Z^1 \subseteq Z^2$, and so on. In the spirit of proposition 7.5, give necessary and sufficient conditions on the sequence of observations being consistent with this sort of model.

■ 7. Suppose that for some reason we were interested in studying solutions to the following problem: A firm produces a single output y from N inputs $x = (x_1, \dots, x_N)$. The firm is given a certain budget that it can spend on inputs, and the firm produces as much output as it can given that budget. Letting B be the budget and supposing the firm is described by a production function $f(x)$ and that the firm is competitive and faces factor prices w , we can define

$$y^*(w, B) = \max\{f(x) : w \cdot x \leq B\}.$$

Devise a theory concerning this problem in the spirit of the results we have given concerning partial derivatives of y^* and solutions to the problem just posed.

■ 8. Carry out the proofs of (1), (2), and (3) that are used in the proof of proposition 7.9.

■ 9. Give results in the spirit of proposition 7.9 for the cost and conditional factor demand functions. If it makes things easier, you may restrict attention to the case of a single output. (If you can do this, you may want the further challenge of giving a proof of the "integrability" result for Marshallian demand.)

■ 10. Show that if the technology of a one-output firm has nondecreasing returns to scale, then average costs for the firm (as a function of the amount of output) will be nonincreasing. Show that if the technology has constant returns to scale, then average costs are constant. Show by example that the technology of a one-output firm may have nonincreasing returns to scale and yet average costs for the firm fall with scale of the output, at least for some levels of output. (All of these are for fixed factor prices and a competitive firm.)

■ 11. Consider the model of the competitive firm with N inputs, one output, and production function f . We say that the firm has *homothetic conditional factor demands* if, for fixed factor prices w , the optimal conditional factor demands do not change proportions as we change the scale of output; that is, if x^* solves $FCMP(w, 1)$, then for every $y > 0$ there is an $\alpha(y) > 0$ such that $\alpha(y)x^*$ is a solution of $FCMP(w, y)$.

(a) Show that the firm has homothetic conditional factor demands if $f(x)$ takes the form $\phi(F(x))$ for a function $F: R_+^N \rightarrow R$ that is homogeneous of degree one and $\phi: R \rightarrow R$ is strictly increasing.

- (b) Which of the production functions in problem 1 give homothetic conditional factor demands?
- (c) If a firm has homothetic conditional factor demands and nonincreasing returns to scale, what can you say about the behavior of average costs as scale increases (for fixed factor prices)?
- (d) Give an example of a firm without homothetic conditional factor demands.

chapter eight

The competitive firm and perfect competition

Now that we have a way to model firms, we can include them in models of price-mediated, market equilibrium. In this chapter we will consider market equilibria for competitive firms, firms that act as price takers.

We move immediately to a caveat: In all the models of this chapter, there will be a finite number of firms and (at least implicitly) a finite number of consumers. Prices will change if there are changes in the aggregate supply and demand decisions of firms and consumers, so it *cannot* be correct that all firms and all consumers do not affect prices by their actions. If there are many firms and consumers, each individual firm and consumer may have very small effect on prices, but the effect is not zero. The theory we develop is not based on a hypothesis that firms and consumers have no effect on prices. Instead it is based on the hypothesis that they *act* as if they have no effect; consumers choose what to consume and firms choose their production plans in the belief that the prices they see are unaffected by their decisions. Consumers and firms, then, act on an incorrect supposition. But if, the story is meant to go, there are many consumers and firms, and if no single consumer is a J. D. Rockefeller and no firm is a General Motors, then these suppositions are only a little incorrect.

There are two ways to proceed in the theory. We could continue analysis of general equilibrium in the style of chapter 6, but with firms added to the story. Or we can undertake *partial equilibrium* analysis — analysis of the market in a single good (or in a few related goods), holding fixed what goes on outside our limited frame of view. We will do both of these, and in doing so we begin an important transition in the style of this book. After the general equilibrium parts of this chapter, we will generally tend to look at quite small pieces of a larger economic picture; our attention hereafter will usually be very focused.

This is not to say that general equilibrium is completely broad and all-encompassing. Even if we have a model that incorporates all markets, market behavior is still embedded in a larger social and political environment,

which general equilibrium takes as given. The difference between our partial equilibrium analyses and general equilibrium analyses is one of degree and not of type. But, as you will see, the difference in degree is substantial.

We begin with the classic partial equilibrium analysis of perfect competition. We then develop an example that shows how a partial equilibrium perspective can be misleading, and with that as stimulus, we briefly discuss general equilibrium with firms.

8.1. A perfectly competitive market

In the standard partial equilibrium analysis of perfect competition, we imagine a number of firms, all of whom can produce a single good, which is sold to consumers in exchange for some "numeraire" good, money. The market operates under the following qualitative conditions.

- (a) The consumers of the good in question are happy to have the good from any of the many producers/vendors of the good. No vendor of the good has any particular advantage in selling to any consumer; no consumer would knowingly pay a higher price to one vendor if the good can be had from another vendor at a lower price. In jargon, the good is *undifferentiated* and/or is a *commodity*.
- (b) The consumers have perfect information about the prices being charged by the various vendors of the good.
- (c) Vendors of the good are willing to sell to any buyer, and they wish to get as high a price as they can. They have perfect information about prices being paid by consumers elsewhere, and they have the ability to "undercut" competitors if it is worth their while.
- (d) Resale of the good in question cannot be controlled, and both sale and resale are costless.

These four conditions are taken to imply that *all exchanges will take place at some single price for the good*. These conditions are also taken to imply that *prices are linear*; there is a single unit price for the good, and if a consumer wishes to purchase n units of the good, he or she pays n times whatever unit price the good commands.

To justify formally that there will be a single price and prices will be linear, we would need to flesh out a precise model of the market institutions. We would have to say what the forum for these exchanges is, how prices are set, and so on. In the theory of perfect competition, this

specification is not done. Rather, the italicized "conclusions" just given are really the assumptions of the theory, and the four things listed as "conditions" are our excuses for these real assumptions. It is held that in real markets where the four conditions and the others that follow hold to some degree of approximation, the conclusions listed above and others that follow will be borne out. This, of course, is an empirical question. For some industries this is true, and for others it is not.¹

We could alternatively imagine that the good in question can be sold at nonlinear prices, in the sense that quantity discounts or premia can be charged to individual consumers and consumers shop for the best price schedule. For this sort of thing to work, it must be possible to control resale of the good. A possible example of this would be telephone services in the United States, where phone services are not offered by a (government or other) monopoly; AT&T, MCI, Sprint, etc., might offer quantity discounts, where "resale" of phone services is not possible if the service is tied to a particular phone in a particular location. Even if resale is not possible, if there is really very good information about the prices the various carriers charge and if consumers don't mind from which carrier they buy phone services, then we can imagine perfect competition in this industry without linear prices. But we won't try that on a first pass through the subject.

To continue with informal conditions:

- (e) Each individual consumer takes price as given. No consumer believes that he or she could in any way change the price paid per item by changing his or her level of demand for the item.
- (f) The producers/vendors of the item in question (which will hereafter be called the firms) also take as given the market price for the good in question. They believe that they can sell as much of the item as they wish at the going price, and they can sell none at all at a higher price.

So we are led to the following:

Definition. *Equilibrium in a perfectly competitive market is given by a price p for the good, an amount purchased by each consumer, and an amount supplied by each firm, such that at the given price each consumer is purchasing*

¹ If you wish to see an industry where the assumptions and predictions of perfect competition can be tested, read about the *Oil Tanker Shipping Industry*, as described in the Harvard Business School Case #9-384-034. (This case is reprinted in Porter [1983], which contains a number of other cases that provide interesting illustrations of concepts discussed in this book.) If you do this, be forewarned that this is an industry in which the model of perfect competition both works and doesn't work. We'll say a bit more about this epigram in a later footnote.

her preferred amount and each producer is maximizing profits, and the sum of the amounts purchased equals the sum of the amounts supplied.

The reader is more than entitled to some befuddlement at this definition. How can we say what a consumer will demand as a function of the price of this one good? How can we say what a firm will supply? Presumably we will use the model of the consumer from chapter 2, but then we need to know what the other prices are and what form the consumer's wealth takes: perhaps an amount of the numeraire good, as in the stories of Marshallian demand in chapter 2; or some endowment bundle, as in chapter 6. As for firms, presumably they will all be described by one of the methods explored in the last chapter. But we will need to say something about the entire array of prices they face.

The consumer side: Demand curves

In most treatments of partial equilibrium analysis, the consumer side of the market is pretty much ignored. In the back of our minds are consumers as in chapter 2 and an assumption that prices of other goods that the consumers might consume are fixed.² And so in the back of our minds is an assumption that we can write, for each price p that the good in question might command, the total amount the consumer side of the market will demand, denoted $D(p)$. All these things stay in the back of our minds, and we simply take as given a demand curve $D(p)$. Moreover, it is typically assumed that this demand curve is downward sloping — more is demanded the lower the price of the good.

Nothing stops us from bringing all this up to the front of our minds. We could present a detailed model of the consumer side of the market. Later, when we look at "competitive" markets in which the goods aren't quite commodities and we become interested in the shape and character of aggregate demand, we will do just this. And we will do this when, for various reasons, we become interested in a more detailed look at market institutions.^a

But the focus in the classic partial equilibrium analysis of perfect competition is on firm behavior, so it is typical to simplify the consumer side and simply posit the existence of a downward sloping demand function.

² But see section 8.3.

^a If you wish to see a bit of this, look ahead to problem 3 in chapter 10. The issue there is one of rationing a limited supply among a number of consumers. Specifically, we ask how the rationing scheme employed for one good affects demand for a "second" good. Since rationing schemes work at the level of individual consumers, we have to disaggregate market demand and consider the preferences of individual consumers and how those preferences and different rationing schemes lead to amounts demanded for the second good.

We will follow this tradition until we get to general equilibrium later in this chapter.

Firms: The supply function

We attend to firms in greater detail. We will assume that each firm produces only the good in question from some array of factors of production. There will be J firms, indexed $j = 1, \dots, J$, and firm j will be described for now by a total cost function $TC_j(y)$. Since firm j acts competitively in the product market, if the equilibrium price of the good is p , firm j supplies the amount $y_j(p)$ that solves

$$MC_j(y_j(p)) = p.$$

And total supply at price p is $S(p) = \sum_{j=1}^J y_j(p)$.

It is typical to assume that total costs are convex, so that marginal costs are nondecreasing and the first-order condition $MC_j(y) = p$ gives an optimal level of production. If total costs are strictly convex, marginal costs are strictly increasing, and the supply of the firm at price p , $y_j(p)$ is uniquely defined and increasing in p .

If this holds for all firms, $S(p)$, hereafter called *industry supply*, is strictly increasing. And now putting together the unmodeled demand side of the market specified by the demand function $D(p)$ and the slightly modeled supply side characterized by the supply function $S(p)$, we have an equilibrium at the price p where $D(p) = S(p)$. One has the standard pictures from intermediate and even principles textbooks shown in figure 8.1. Note well one graphical characterization developed there: The industry supply curve is the horizontal sum of individual firms' marginal cost curves.

8.2. Perfect competition and -runs

The short and intermediate runs

If the story stopped there, we wouldn't have much of a story. So we proceed to elaborate the supply side story.

(1) We imagine that the firms are characterized by production functions plus a fixed cost. If firm j is in the business of producing this good, it must pay a fixed cost K_j . It must pay this cost even if it chooses to produce zero output; this cost is avoided only if the firm leaves the industry, an option firms will not be given for now. And if the firm is in business, it produces output from various combinations of N factor inputs, where

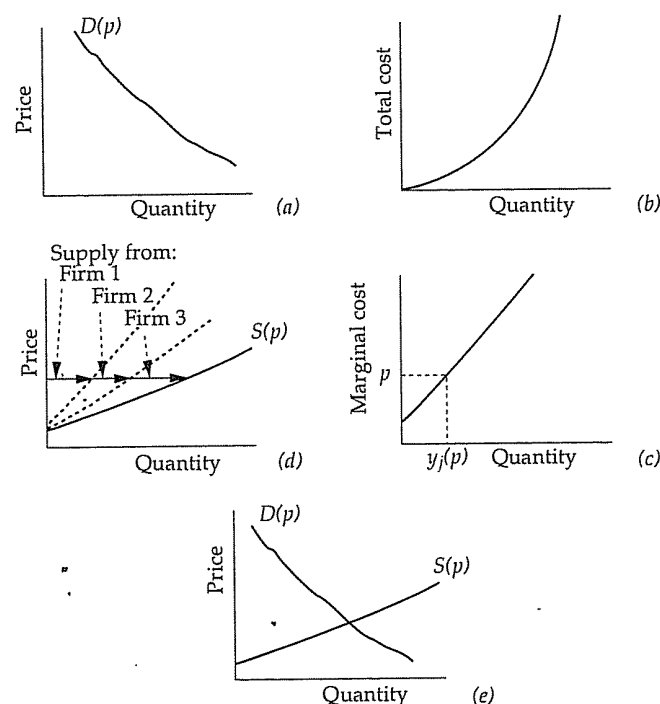


Figure 8.1. Equilibrium in a perfectly competitive market.

Demand is given in the form of a downward sloping demand curve, as depicted in (a). Each firm has a total cost curve as in (b) and a corresponding marginal cost curve as in (c). At each price p , firm j supplies the amount $y_j(p)$ that equates marginal cost with price (shown in [c]). Industry supply is obtained as the horizontal sum of individual firms' supply functions, which is just the horizontal sum of individual firms' marginal cost functions (shown in [d]). Equilibrium is where supply (computed in [d]) intersects demand (given in [a]). Superimposing (a) and (d), we get (e) and the equilibrium price and quantity.

$f_j(x_1, \dots, x_N)$ denotes the production function, the amount of output derived from factor inputs $(x_1, \dots, x_N)$. Note well: The total costs of firm j if j uses the factor inputs $(x_1, \dots, x_N)$ is the sum of costs of those inputs plus the fixed cost K_j .

(2) We imagine that the firms are price takers not only in the market for

their output but also in the factor markets, and we let $w = (w_1, \dots, w_N)$ be the vector of factor prices, which is taken as fixed throughout our analysis.

(3) We imagine that firms begin with some initial levels of the factor inputs, some of which can be adjusted in the "short run," while others can only be adjusted in the "intermediate run."

Breaking time in this fashion, into a short run and an intermediate run and then partitioning the factor inputs into those that are short-run adjustable and others that are intermediate-run adjustable is hardly realistic. It would be more realistic to suppose that some of the firms can change the levels of all their factor inputs quickly, while others may be able to change very little until a lot of time has elapsed. Still, we explore what will happen in this market in the short and in the intermediate run, assuming that these time frames are meaningful and apply to all firms in the industry, and assuming that what is fixed in the short run for one firm is fixed in the short run for all.

We assume that factor inputs $n = 1, \dots, N'$ are fixed in the short run and variable in the intermediate run, and factor inputs $n = N' + 1, \dots, N$ are variable in both the short and the intermediate run. We also assume that the industry starts off at a point where firm j has been using factor inputs at the levels $(\hat{x}_1^j, \dots, \hat{x}_N^j)$.

What, then, does industry supply look like in the short run? In the short run, the total variable cost function of firm j , written $SRTVC_j(y)$ is

$$\min_{(x_{N'+1}, \dots, x_N)} \left\{ \sum_{n=N'+1}^N w_n x_n : y = f_j(\hat{x}_1^j, \dots, \hat{x}_{N'}^j, x_{N'+1}, \dots, x_N) \right\},$$

and the short-run total cost function is

$$SRTC_j(y) = SRTVC_j(y) + \sum_{n=1}^{N'} w_n \hat{x}_n^j + K_j.$$

The short-run marginal cost of firm j is the derivative of the short-run total cost (which is the same as the derivative of the short-run total variable cost), and short-run industry supply is the horizontal sum of the firms' short-run marginal cost curves, just as in the previous section.

What does it take for $SRTC$ to be strictly convex?

Proposition 8.1. *If the production function f is strictly concave, then the total cost function (for fixed factor prices, for any run as long as some of the factors are variable) is strictly convex.*

This is left for homework.

In the intermediate run, the total variable cost function of firm j is

$$IRTC_j(y) = \min_{(x_1, \dots, x_N)} \left\{ \sum_{n=1}^N w_n x_n : y = f_j(x_1, \dots, x_N) \right\} + K_j.$$

This gives intermediate-run marginal cost curves, and the intermediate-run industry supply curve is the horizontal sum of these marginal cost curves.

How will short-run and intermediate-run supply curves compare? The easiest way to think about this is to begin with the relevant short- and intermediate-run total cost curves. In the short run, where the firm can adjust only variable factors of production, the total cost of production will be everywhere at least as much as intermediate-run total cost. As long as the short-run fixed factors of production are at an optimal level for some level of output, the two total cost curves will touch at that level of output. We're assuming rising marginal costs, so total costs are convex; the picture is as in figure 8.2(a). And now marginal cost is the slope

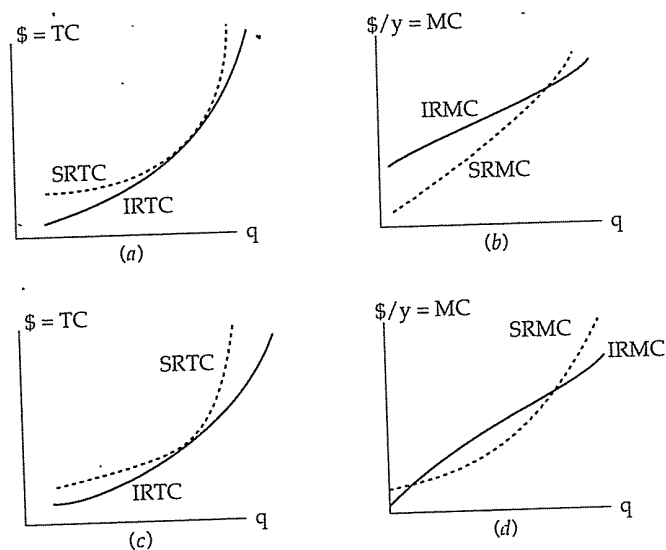


Figure 8.2. Short- and intermediate-run total cost curves and marginal cost curves.

of total cost, so that the two marginal cost curves are as in figure 8.2(b). Note well: This has been drawn so that short-run marginal cost is lower than intermediate-run marginal cost for all low levels of output and higher for all high levels of output. But this particular drawing of the two isn't necessarily implied by figure 8.2(a). In figures 8.2(c) and (d) another case is given. What is implied in general is that short-run marginal cost is lower than intermediate-run marginal cost for quantities just below the point where the two total costs are tangent and is higher at quantities just above.³

This tells us how supply changes from the short to the intermediate run. What about demand? There are certainly good reasons to think that demand in the short-run will be different from demand in a longer intermediate-run time frame. If, to take one case, the good in question is a variable factor of production in some downstream production process, then short-run demand for it will be based on calculations that do not allow producers to substitute it for (or substitute for it) more-fixed factors in downstream production. In the intermediate run, substitution of/for more fixed factors may come into play. We'd expect demand to shift according to the run in such cases.⁴ Even if the good in question is a final consumption good, we can think of reasons that short- and intermediate-run demand will be different. If the good is somewhat durable, then a price shift, say a rise, will likely reduce demand more in the short run than it will in the longer run. (Why?) If the good is an addictive good, the reverse is true. (Why?) Despite these examples, it is typical in the theory to work with the case where market demand doesn't change with changes in the time frame, and we will henceforth make that assumption.

The long run

We continue to elaborate the theory by adding a long-run time frame. We begin with J firms producing the good in question. But we imagine that these J firms are not committed forever to continue producing the good, and we imagine that there are many potential producers of the good in question. In the long run, firms can enter and/or leave this industry: Firms that are in the industry are assumed to depart if at equilibrium prices they are sustaining losses. And firms will enter the industry if at equilibrium prices they can make strictly positive profits.

³ Can you prove this?

⁴ Admittedly, this is "cheating" a bit, since we didn't mention in our discussion of the demand side of the market the possibility that demanders are other firms. But there are certainly markets in which this case would pertain.

Note well: Firms that can at best make zero profits are just on the cusp of the entry/exit decision. A firm that is producing and makes zero profits will be willing to remain, whereas a firm that is not producing and would make zero profits at the going equilibrium price is assumed to stay out.^b

We formalize this elaboration as follows. We imagine that many firms could be in this business, indexed by $j = 1, \dots, J^*$ for a very large J^* . Each firm is characterized by a production function f_j and a fixed cost K_j , as stated. Then,

Definition. A long-run equilibrium in a perfectly competitive market consists of (a) a price p for the good, (b) a list of active firms taken from the list of all potentially active firms, and (c) for each active firm, a production plan such that

- (d) Each active firm is maximizing its profits, taking prices as given, at the production plan prescribed in (c)
- (e) Each active firm is making nonnegative profits with its production plan
- (f) Each inactive firm would make nonpositive profits at best, taking prices as given, if it became active
- (g) The total supply from the active firms, gotten by summing their output levels as specified by their production plans, equals market demand at the equilibrium price p

Note that this is just a specialization of our general definition of equilibrium in a perfectly competitive market, if we think that the long-run total costs for each firm are zero at zero level of production and rise to K_j if any production at all is undertaken.

It does seem rather a mess, but we make a fair bit of progress in clearing up this mess if we add one final assumption.

Assumption $\star$. For every firm that is potentially active in this business, there are "many" other potentially active firms that have precisely the same production function and fixed costs.

^b Recall the earlier epigrammatic comment about the model of perfect competition holding and not holding in the oil tanker shipping industry. With regard to this, note that firms might behave in the way the theory says they do in the short and intermediate run, and at the same time behave quite crazily from the perspective of the theory in the long run. Then short- and intermediate-run predictions of the theory could be borne out, while long-run predictions, which we are about to develop, would fail.

Informal Proposition. If assumption $\star$ holds, then in any long-run equilibrium, all active firms make precisely zero profits.

We call this an informal proposition because it is based on the very informal "many" in the assumption. The logic runs as follows: For any active firm in a long-run equilibrium let π be its level of profits. If there are "many" firms like this one, some of those will be inactive. *This is the sense of "many"; it must be that for every active firm at least one firm identical to it is inactive.* In a long-run equilibrium, all inactive firms must be capable of making zero profits at best. Since there is an inactive firm identical to our active firm, it could (if it entered) make profits π . So π must be nonnegative, if the active firm is to stay active. And π must be nonpositive, if the inactive firm is to remain inactive, which implies that π must be zero.

Suppose that an active firm is, in fact, making zero profits at the equilibrium price p . Consider where it is producing along its long-run average cost curve. It must be producing at its point of minimum long-run average cost, and this minimum level of long-run average cost must be the equilibrium price. To see this, first note the accounting identity

$$\text{Profit} = [\text{Price} - \text{LRAC}(\text{Quantity})] \times \text{Quantity},$$

where LRAC is the long-run average cost function. Now if price exceeds the minimum level of LRAC, the firm, by producing at the level of minimum LRAC, would make positive profits. If price is less than the minimum level of long-run average costs, the firm cannot make nonnegative profits at any level of output. The only way that a profit-maximizing, competitive firm can make zero profits is if price equals minimum LRAC. And the firm then makes zero profits only by producing at a quantity that minimizes LRAC.

This discussion leads immediately to the following corollary to the informal proposition.

Informal corollary. If assumption $\star$ holds, then in any long-run equilibrium every firm is producing at efficient scale, where efficient scale is any scale that minimizes the average costs of the firm.

Assumption $\star$ is often made part of the defining characteristics of a perfectly competitive industry. The justification for it is that in a competitive industry, no firm has a technological advantage over any other — whatever technologies there are will be freely available to anyone who cares to use them. Hence we always will have potential entrants identical to anyone who is active in the business.

And once we have assumption $\star$, we arrive at the conclusion: In a long-run equilibrium, firms make zero profits and produce at efficient scale.

Of course, the rest of the assumptions of the theory hang together very nicely if we don't make assumption $\star$. Think, for example, of the wheat farming industry. We might well consider wheat to be a commodity good with a single, well-known price, which all wheat farmers and wheat buyers take as given. Still, some wheat farmers are lucky enough to own land that is especially good for growing wheat; these farmer-firms have a "technology" that is superior to the technology owned by farmer-firms who farm less fertile land. So we expect those farmers to make strictly positive profits.

When we don't make assumption $\star$, so it is possible that firms can make positive profits in a long-run equilibrium, we also don't expect those firms to be producing at efficient scale. In fact, they won't do so; these firms will always produce at larger-than-efficient scale. You are asked to prove this in problem 2.

In such cases, one can (and some authors do) define things so that, even though assumption $\star$ is not made, profits still equal zero in a long-run equilibrium. This is done by saying that what a firm with a "productive advantage" earns are not profits, but rather *rents* to its advantaged production capabilities. These rents come off the top as costs to be paid for the advantaged technology, so that the firm is back to making zero profits. (If you like this sort of proof-by-definition, consider why this means that the firm now is producing at the point of minimal LRAC.)

An example

To clarify some of the points made, consider the following simple example. The numbers in this example are selected because they give relatively clean and simple answers; they are not meant to be at all realistic.

In a particular economy, a product called *pfillip*, which is a nonnarcotic stimulant, is produced by a competitive industry. Each firm in this competitive industry has the same production technology, given by the production function

$$y = k^{1/6} l^{1/3},$$

where y is the amount of *pfillip* produced, k is the amount of *kapitose* (a specialty chemical) used in production, and l is the amount of *legume* (a common vegetable) used in production. Firms also incur fixed costs of \$1/6.

Legumes are traded in a competitive market, and the price of a unit of legume is a fixed \$1, regardless of the amount of legume demanded for the production of *pfillip*. The level of legume can be freely varied in the short run. Kapitose is traded in a competitive market, at price \$1/2. The amount of kapitose used by any firm in production cannot be varied in the short run but can be adjusted in the intermediate run.

Many firms could enter this industry in the long run, and firms are free to depart. All firms, both those in the industry and potential entrants, have the technology and cost structure just described.

Demand is given by the demand function $D(p) = 400 - 100p$, where p is the price of *pfillip* and $D(p)$ is the amount demanded at this price.⁵

What is the long-run equilibrium in this perfectly competitive market?

As a first step, let us compute the total cost function of each firm in this industry, where we assume that the firm can vary both its level of kapitose and legume inputs.

The cost-minimizing way to produce y units of output is the solution of

$$\min_{k,l} \frac{k}{2} + l, \text{ subject to } k^{1/6} l^{1/3} \geq y.$$

It is clear that the solution will have the constraint binding. (We should put in nonnegativity constraints, although they will not bind and so we've left them out.) Since the constraint will bind, we can solve for k in terms of l to obtain $k = y^6 / l^2$, and so the problem becomes

$$\min_l \frac{y^6}{2l^2} + l.$$

The first-order condition is

$$\frac{y^6}{l^3} = 1, \quad \text{or} \quad l = y^2 \quad \text{and} \quad k = y^2,$$

⁵ We will use models with linear demand throughout this book, and it should always be understood that demand is nonnegative; that is, $D(p)$ is really $\max\{400 - 100p, 0\}$ in this example. We also should be careful about demand at a price of zero; what is important is that revenue at a price of zero is zero.

so that the long-run total cost for producing y is

$$TC(y) = \frac{3}{2}y^2 + \frac{1}{6}.$$

From this we can easily find the long-run equilibrium price: It must be the minimum value of average cost. Average costs are given by $AC(y) = (3/2)y + 1/(6y)$, which is minimized at $3/2 = 1/(6y^2)$, or $y = 1/3$. That is, producing $1/3$ of a unit is the efficient scale for firms in this industry. And when firms produce $1/3$ of a unit, they have average costs of \$1. So the long-run equilibrium price is \$1.

At this price, demand is $400 - 100(1) = 300$, so total industry demand is 300. And since each firm will be producing $1/3$ of a unit, there will have to be 900 active firms.

To summarize, at a long-run equilibrium 900 firms will be active, each producing $1/3$ of a unit, for a total industry supply of 300 units. The equilibrium price will be \$1, and each firm will be making zero profits. Each of the 900 firms will be utilizing $1/9$ unit each of legume and kapitose.

Now suppose the demand curve suddenly shifts, becoming $D(p) = 750 - 150p$. What will be the industry response in the short run, in the long run, and in the intermediate run?

We will take the short run first. In the short run, there are 900 firms, each of whom is fixed at $1/9$ units of kapitose. If any of these firms wishes to have output level y , they must employ enough legume l so that

$$(1/9)^{1/6}l^{1/3} = y \text{ or } (1/9)l^2 = y^6 \text{ or } l = 3y^3.$$

This gives total variable costs of $3y^3$. Hence the short-run marginal cost function of the firm is $MC(y) = 9y^2$. And so, at price p , a single firm supplies the amount $y(p)$ that solves $9y(p)^2 = p$ or $y(p) = \sqrt{p}/3$. Since there are 900 identical firms, industry supply at price p is 900 times this, or $S(p) = 300\sqrt{p}$. The short-run equilibrium price is where short-run supply equals demand:

$$300\sqrt{p} = 750 - 150p.$$

If you solve this equation for p , you get (approximately) $p = 2.1$.

The short-run equilibrium price is \$2.10. At this price, total industry demand is $750 - 150(2.1) = 435$, which is divided among the 900 firms so that each firm produces approximately .483 units. This requires each firm to employ approximately .338 units of legume. Each firm must also pay the cost of the fixed $1/9$ unit of kapitose and the $1/6$ fixed cost, for total costs of $.338 + (1/9)(1/2) + 1/6 = .56$ against which each firm makes revenue of $(2.1)(.483) = 1.014$, for a profit per firm of .454.

Moving to the intermediate run, we still have 900 firms active, but now each has total cost function $(3/2)y^2 + 1/6$, or marginal costs $MC(y) = 3y$. Hence at price p , each firm supplies (in the intermediate run) $p/3$ units, and the intermediate-run industry supply curve is $300p$. Intermediate-run equilibrium is where $300p = 750 - 150p$ or $p = 5/3 = 1.667$. So we conclude:

The intermediate-run equilibrium price is \$1.667. At this price, total industry demand and supply equal 500 units, and each firm produces $5/9$ of a unit. This costs the firm $(3/2)(5/9)^2 + 1/6 = 34/54$, against revenues of $(5/3)(5/9) = 25/27$, for a net profit of $16/54 = \$296$. (We leave it to you to work out kapitose and legume utilization per firm.)

And the long run is easiest of all. We already know that at the long-run equilibrium price must be \$1 and each firm must produce $1/3$ of a unit. So,

The long-run equilibrium price is \$1, which means that industry demand is for 600 units. Each firm produces $1/3$ of a unit, so the number of firms doubles to 1,800. Each firm makes zero profits, and each firm utilizes $1/9$ unit each of legume and kapitose.

We draw these "dynamics" in figure 8.3. We show there the original demand curve and the long-run supply curve in solid lines. Note that the long-run supply curve is horizontal as it must be in this sort of industry. The short-run and intermediate-run supply curves beginning from the initial equilibrium position are drawn in as dashed lines; note that the short-run supply curve has the steepest slope. We also draw in the "new" demand curve as a dashed line, and we mark the stages of industry response to the shift in demand: (1) marks the original equilibrium, (2) the new short-run equilibrium, (3) the new intermediate-run equilibrium, and (4) the new long-run equilibrium. Note that equilibrium quantities rise at each stage, and prices jump way up and then move down. Also note (not from the picture) the pattern of kapitose and legume usage by one of the originally active 900 firms. In the short run legume usage rises, then

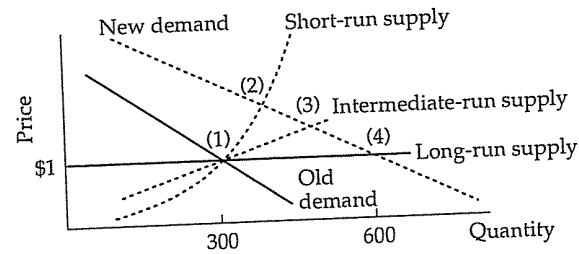


Figure 8.3. Equilibrium "dynamics" for a perfectly competitive industry.

it falls in the intermediate and long runs. Kapitose usage per firm stays constant in the short run (as it is constrained to do), then rises, and then falls.

The qualitative features of this simple example generalize substantially: In an industry where assumption $\star$ concerning potential entrants holds (and factor prices don't change with the scale of industry output; cf. the next section), long-run supply will necessarily be perfectly elastic (flat), so shifts in demand will not change the long-run equilibrium price for the good. Moreover, as asserted in section 8.2, short-run supply will generally be more inelastic than intermediate-run supply, at least for small changes in quantity, for a fixed number of firms. Hence the qualitative features of figure 8.3 can be expected to hold generally (if the story about all these runs is correct); an upwards shift in demand will be met in the short run by a small increase in quantity and a large increase in price, in the intermediate run by a further increase in quantity and a decrease in price (still above the original level), and in the long run with price retreating to the original level and output (and the number of firms) expanded to meet this increased demand.

Other changes can lead to other dynamics. For example, in problem 6 you are asked to sketch out what happens if one of the factor prices suddenly changes. Those dynamics can also be generalized beyond simple parametric examples, at least in cases where cost and demand curves are assumed to be "typical."

Such exercises illustrate the formidable power of partial equilibrium analysis in general and partial equilibrium analysis of perfect competition in particular. If cost and demand curves are "typical," which is to say demand is downward sloping and supply is nondecreasing, then the theory predicts a unique equilibrium. (Compare with general equilibrium. And, when we get to it, compare this with the predictions made using the methods of part III.) Insofar as we can empirically estimate demand functions

and cost functions, we can construct the theoretical industry supply curve and get very good tests of the theory. Moreover, this runs version of the theory makes very definite predictions about how equilibrium prices and quantities will respond dynamically to shifts in parameters of the model, again giving us a lot of empirical leverage. While our example is only an example, it indicates how powerful the theory can be.

8.3. What's wrong with partial equilibrium analysis?

At the same time, the theory is predicated on a number of assumptions that may not prove true, particularly assumptions that are required by the narrow focus of a partial equilibrium model.

For example, in a partial equilibrium analysis of the sort we have been conducting, we hold "fixed" various things left out of the model. It is typical in a partial equilibrium analysis of the market in a single product to say that one is holding fixed the prices of all commodities not in the market. For example, if we were more explicit about the demand side of our markets, we would look at consumer demand functions in the spirit of chapter 2. And it would be typical to analyze how demand for the commodity in question changes, holding fixed the price of other commodities and holding fixed the incomes the consumers have to spend. Rendered in symbols, if $x_k^i(p_1, \dots, p_K, Y^i)$ is the demand function of consumer i for good k as a function of all prices and the consumer's income Y^i , the usual practice is to think of the industry demand curve as arising from fixing all prices except p_k , say at levels $\hat{p}_k$, fixing all the Y^i , and then, for each price p_k for good k , saying that

$$D(p_k) = \sum_{i=1}^I x_k^i(\hat{p}_1, \dots, \hat{p}_{k-1}, p_k, \hat{p}_{k+1}, \dots, \hat{p}_K, Y^i),$$

where $i = 1, \dots, I$ in the sum indexes the set of consumers.

In many applications, this would be the wrong thing to do, if you were really interested in industry demand. Suppose the commodity being considered is wheat. Of course, the demand for wheat depends on other factors, such as the price of corn. Should we write down the demand curve for wheat, holding fixed the price of corn? This is what is typically done and what the scheme above advises. But is it the right thing to do? As the price of wheat changes, the level of demand for corn will change. It is natural to expect that demand for corn rises as the price of wheat rises. Unless the supply of corn is perfectly elastic (is perfectly flat), the price of

corn shifts. And this then will change how much wheat will be demanded. In the short run at least, it makes little sense to suppose that the supply of corn is perfectly elastic. And so it makes little sense to suppose that the demand curve for wheat, at least in the short run, should be computed on the basis of an unchanging price of corn. *Changes in the price of wheat will, through the workings of markets, change the price of corn, which then affects the demand for wheat.* Predictions on the demand for wheat (as the price of wheat changes) that are based on a fixed price of corn will, therefore, be wrong.

If you are analyzing the market in a commodity where changes in its price (and the corresponding level of demands) won't much change the prices of other goods, then a demand curve computed under the hypothesis that other prices don't change at all won't be too far wrong. But unless this condition holds for the commodity you are interested in, you will want to think about whether there might be other prices that move substantially as the price of the good in question moves, with feedback to the demand for the good in question. If other prices can reasonably be expected to move with movements in the price of the good in question, then you must analyze more than the single market, if reasonable predictions are to emerge.

The same sort of consideration arises on the supply side of the market. Our analysis in section 8.2, for example, was predicated on an assumption that the prices of factors of production don't change. We might think to justify this assumption by appealing to the notion that many firms are in the industry, and so each is a price taker in all the factor markets. But this is no justification at all. As we change industry supply levels, we aren't looking at changes in demand for a given factor caused by a single firm; instead we are looking at changes in demand for the factor caused by changes in the activities of many firms in the industry. If this factor of production is used by many firms in the industry, if demand for this factor by firms in the industry is a large part of the demand for the factor, and if supply of the factor is not perfectly elastic, then it is wrong to build industry supply curves based on the supposition that the prices of the factors don't change with changes in the level of industry supply.

An elaborate example

Let us illustrate by continuing with the example from the previous section. We will continue to suppose that the price of legume is \$1 per unit, no matter how many units of legume are bought by producers in the pfillip industry. But we suppose that the price of kapitose changes with changes in pfillip industry demand for kapitose. In particular, we suppose that if the

pfillip industry demands K units of kapitose, the price per unit that kapitose will command is $K/200$.

Why might this happen? We might suppose that kapitose is produced by an industry of perfectly competitive firms, but entry to this industry is blocked on legal grounds — say only a given set of firms are licensed to produce kapitose. If entry to the industry is restricted for this or any other reason, then even when firms in the industry are price takers, the industry supply curve may be rising; see problem 4. Or the kapitose industry might be perfectly competitive and with free entry, but potential entrants have progressively less efficient technologies for producing kapitose; see problem 5.⁶

When kapitose was assumed to trade at \$.50 per unit regardless of demand for kapitose from pfillip producers, the long-run supply curve of pfillip was perfectly elastic at price \$1. We now proceed to compute the long-run supply curve of pfillip under the newly presumed conditions. This gets a bit involved, but try to persevere.

Suppose the price of kapitose is q at some point. Each individual pfillip producer is (assumed to be) a price taker, so each computes long-run total costs of producing y units, given this factor price, as

$$\min_{k,l} qk + l + \frac{1}{6} \text{ subject to } k^{1/6}l^{1/3} \geq y.$$

If you work through the math, you will find that this leads to a total cost function

$$TC(y) = \frac{3}{2}(2q)^{1/3}y^2 + \frac{1}{6},$$

and an average cost function

$$AC(y) = \frac{3}{2}(2q)^{1/3}y + \frac{1}{6y}.$$

Efficient scale for a firm with these average costs is the solution of $(3/2)(2q)^{1/3} = 1/(6y^2)$; if you manipulate the algebra, you will find that this is

$$y^e(q) = \frac{1}{3(2q)^{1/6}},$$

where the superscript e stands for *efficient*. At this efficient scale, the level of average costs is

$$AC^e(q) = \frac{3}{2}(2q)^{1/3} \frac{1}{3(2q)^{1/6}} + \frac{1}{6/[3(2q)^{1/6}]} = (2q)^{1/6}.$$

⁶ Yet another possibility is that the kapitose industry is not competitive at all. In the problems at the end of chapter 9, you will be asked to consider the case of a monopoly kapitose producer. This situation is harder to deal with because the notion of a kapitose supply curve is not well defined in a monopoly situation.